Behavioral responses to fines: direct and indirect deterrence

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Abstract

Governments use fines to deter socially unwanted behavior on a wide array of issues. It is commonly assumed that, fixing the probability of sanction, the higher the price, the more deterrent a fine is. But with limited government capacity for effective enforcement, the fine is often only an invitation for citizens to change behavior. Thus, the perceived fairness of the fine price might moderate deterrence. I explore this behavioral mechanism with data from automated speed cameras in Bogotá, Colombia. I use the fact that the speed threshold for issuing a ticket differs from the posted speed limit (and is unknown to drivers) as a source of exogenous variation, and the high visibility of speed cameras as a way to differentiate direct and indirect deterrence. A regression discontinuity design is used to compare drivers whose speed just exceeded the unknown real limit, and thus get a ticket, with those driving just below the real limit, who do not get a ticket. I find that a ticket causally changes the direct behavior of drivers: ticketed drivers reduce all forms of speeding (direct deterrence). But there is no change in the probability of crashes (indirect deterrence). Using the commercial price of the vehicle as a proxy for wealth, I can also measure the heterogeneous effect of relative price. Drivers at higher wealth levels both reduce speeding and crash participation, while drivers on lower wealth levels show no change in speeding, but an increase in crash participation. My findings are consistent with fines being an invitation by the government to broadly change behavior that is only accepted if it is perceived to be fair. This has specific public policy implications: speeding tickets might not be a well targeted policy for preventing road fatalities. Contrary to extended belief, in some contexts lower fines might be a more behaviorally effective policy tool.

JEL Classification Codes: K42, D91, R41, D61, D62.
Keywords: fines, deterrence, direct deterrence, indirect deterrence, relative price, behavioral responses, traffic crashses, speeding.

1 Introduction

A central question for governments is how to improve the deterrent effect of their legal sanctions. A common assumption is that increasing the price of monetary sanctions results in a higher expected cost for citizens and thus in lesser prevalence of the unwanted behavior (Becker, 1974). But in context where the State capacity for enforcement is limited, fines are also an invitation to change behavior even when the law cannot be enforced. In this later case, citizens accept or reject that invitation. This depends positively on the perception of fairness of the fine: the higher the amount the less fair a fine is perceived to be. Thus, higher monetary sanctions have opposite deterrent effects when the State can and cannot enforce the law in the future.

This paper tries to answer a general effect question: What are the direct and indirect deterrent effects of fines? But most importantly it also explores a heterogeneous effect question: Do these effects vary depending on the relative price of the fine? When talking about relative price I am referring to the fine price relative to personal wealth, as a proxy for how fair the fine is considered to be. Take taxes as an example: the government fines citizens when they lie with their income tax, but it also wants them to pay all other taxes. In contexts with little State capacity for enforcement citizens might know that the State can only audit them on their income tax. Thus, the general effect of the fine can be more negative than expected if frustrated citizens start evading all other taxes. The final behavioral result will depend on the amount of the fine, and the relative weight of income and other taxes.

Speeding tickets in Bogotá, Colombia are a good case study for this question. Speeding tickets are mostly automated in certain locations with road signs and clear visibility [1]. Citizens know where speed is enforced and also know that in the rest of the city speed is not enforced. Thus, future speeding is a good measure of the direct deterrent effect of fines. Since speed is considered the main cause of crashes, exploring future crashes is a good way of measuring the indirect effect of the fine, speeding when the government cannot enforce speed limits. In addition, speed fines in Colombia have a fixed price [2], thus comparing it to a measure of wealth for drivers gives me a good relative price of the ticket. This variable helps me explore the heterogeneous behavioral effects of the fine price.

Examining the effect of speeding tickets on behavior is difficult because of selection.

[1] This has to do with the State stating that it does not want to surprise drivers in any way. A few month before my sample study (May 2024) there was a public campaign to make Cameras and speed signs even more visible. Camera locations are also visible in most transport apps like Waze.

[2] A very important element behind the behavioral model's intuition is the fact that fines in Bogotá are comparatively expensive. The fine is determined by the minimum monthly salary. For all drivers in my data a speeding fine is equivalent to half a monthly salary, or the equivalent of 80 working hours. For comparison, in New York city a speeding fine ranges from 45 to 600 dollars (from 3 to 36 working hours). A higher price will have to be paid in the United Kingdom, where the fines go from £100 to £1,000 (or the equivalent of 8.7 to 87 work hours). A city like Buenos Aires has a much closer price, with speeding fines ranging from \$217.800 to \$1.452.000 Argentinian pesos (from 0.73 to 4 times the monthly salary of 296.832 Argentinian pesos)

Drivers that decide to exceed speed limits might be significantly different in unobservable characteristics that also relate to risky behavior. But in the case of Bogotá, the administrative process for giving speed tickets offers a way to solve this issue. Speed tickets are not given to all vehicles that exceed the maximum speed posted on road signs, but to a different speed with an additional tolerance. Since this hidden threshold is not known to drivers, a discontinuity exists. I use a regression discontinuity design to make a valid comparison between only drivers that decide to exceed the speed limit, but some receive a ticket, and some do not. This allows me to interpret this as a causal estimate. This methodology is confirmed with I a judge-IV on the traffic agent that does a manual verification of each ticket. This alternative estimation confirms my general findings.

I find that speeding ticket have a general direct deterrent effect, measured by the probability of speeding. This effect is considerable in magnitude both at the extensive (77% reduction in the change of reoffending) and intensive margins. I do not find a significant effect on the indirect deterrence, measured with the probability of being in an crash as a proxy for speeding behavior where citizens know that the law is not being enforced. For a subsample I then use the price of the vehicle as a proxy for wealth, to explore my heterogeneous effect question. I find that the direct deterrent effect of speed reduction is explained mostly by the quintiles 4 and 5, and while negative in sign statistically zero for lower quintiles. I then show that the null effect on indirect deterrence for the full sample is really the combination of an increase in the probability of an crash for quintile 2 and a decrease on the probability of a crash on quintile 5. Higher relative price does make fines less behaviorally efficient.

Road crashes are a major public health issue. Road traffic crashes are the leading killer of children and youth aged 5 to 29 years and are the 12th leading cause of death when all ages are considered (World Health Organization, 2023). While this paper tries to answer a more general question on the effect of fines on deterrence, the area in which it answers it can also give actionable and needed policy recommendations.

This paper contributes to four different strands of the economic literature. First, it relates to a wide literature on the intended and unintended effects of State policies on behavior. An important branch of the literature studies what citizens learn by interacting with law enforcement. Using a change on the punishment for recidivism, (Philippe, 2024) finds that learning about criminal law appears to be limited to people with direct involvement with the law. In the same direction, it has been shown that individuals do adjust their priors after receiving a speeding ticket (Dušek & Traxler, 2022). The question of whether a fine is just like any other price (Gneezy & Rustichini, 2000), and what type of price it is (Feldman & Teichman, 2008) has been explored in many dimensions. Another branch of this literature asks whether fairness considerations affect the efficiency of some policies. Taxation is a topic in which fairness considerations have shown to affect behavior in ways that are meaningful for policy design (Best et al., 2025; Perez-Truglia et al., 2024). This paper tries to bring questions of fairness (Khaneman et al., 1986) to the economic understanding of crime, using speeding tickets only as a data-rich case. I contribute to this literature by exploring how interactions with the law affect future behavior when citizens know the law is enforced (direct deterrence) but also when they know is not enforced (indirect deterrence), and with precise estimations at the individual level.

This paper also relates to the specific literature on the analysis of the effects of financial fines and fees imposed by the State. There is ample public debate on the capacity of higher fines to increase revenue and the effect this has on recidivism. Some papers find a modest effect on revenue at the cost of higher recidivism (Giles, 2023). Others argue that even if the effect of sanctions is higher revenue, they have no effect on recidivism and thus are not justifiable ways of increasing deterrence (Finlay et al., 2024). The authors argue that the null effect on all relevant criminal behaviors is precisely estimated, not to lack power. In recent work, (Norris & Rose, 2024) argue that reducing legal financial obligations (LFOs) will impact municipal revenues without increasing deterrence. The evidence provided by (Bing et al., 2024) on a randomized payment of existing fines for criminals shows that this policy only reduces recidivism related to the financial costs of the LFO itself. These effects are not negligible and point to no deterrent effect at a significant cost of engagement with the criminal system. I contribute to this literature by looking beyond recidivism and exploring a behavioral mechanism, perception of unfairness, as a possible explanation.

The question of the optimal amount of a fine has also been examined in the specific case of traffic tickets, but mainly through the lens of revenue, not deterrence like I do here. Some have shown that increasing fines has only a modest effect on timely payment (Traxler & Dušek, 2023) and recidivism (Dušek et al., 2022). A very detailed of the effect of the price of ticket fines is (Kaila, 2024), who uses the relative to income price of traffic tickets in Finland to shows that there is in fact a reduction on recidivism for higher fines, but the effect is short lived and the fines need to be significantly more expensive. In their work, (Goncalves & Mello, 2024) also study the effect of harsher sanctions by exploiting the discretionary sanctions given by police officers. They find that harsher fines do reduce recidivism and, to a lesser degree, crash involvement, but in the aggregate discretion introduces inefficiency. This paper explores the opposite side of the same mechanism; fixed prices fines that in practice are different depending on individual wealth.

Finally, this paper also contributes to a broader literature on the effect of traffic enforcement on behaviors (Angelo & Hansen, 2014; Hansen, 2015; Luca, 2015; Makowsky & Stratmann, 2011). In particular, my works add to a growing literature that studies fines and enforcement in relation to factors that do not affect either fine amount or probability of enforcement but have been shown to have an effect on traffic crashes. Things like the political cycle (Bertoli & Grembi, 2021), or the stock market (Fry & Farrell, 2023; Giulietti et al., 2020). Close to this strand is a set of papers that explores behavioral responses to traffic enforcement and tries to derive policy recommendations to reduce road crashes and fatalities (Lu et al., 2016; Zhang et al., 2020). My work contributes by suggesting perceptions of fairness of sanctions as a mediator of changes in enforcement effectiveness. I also contribute to the traffic literature by adding evidence from a city in the developing world, something that is still rare in the field.

The paper proceeds as follows. Section 2 presents the institutional background of speed fines in Bogotá, Colombia and the data that will be used on the estimation. Section 3 discusses the estimation strategy and presents the evidence necessary for its validity. Section 4 presents all the results from the estimation on the general effect of speeding tickets with a fixed price. Section 5 presents results on the relative price of vehicles. Section 6 presents some robustness checks on the main estimation. Section 7 briefly discusses other behaviors that reinforce the main finding. Finally, section 8 discusses the main findings in relation to fines as an effective deterrent strategy and derives possible policy recommendations for its improvement.

2 Institutional background and data

Automated enforcement of speed started in Bogotá in 2020, with the staggered installation of Cameras in the most dangerous zones [3]. The program was originally called Cámaras Salvavidas (Spanish for Lifesaving cameras) to emphasize that the objective of the program was not municipal revenue but behavioral change. After some debate on the legality of the fines, Colombia's higher court allowed for the fine to be given to the owner of the vehicle [4]. Two elements of the legal frame are crucial in the rest of the paper. First, tickets have a fixed price independent of the type of vehicle or the wealth of the owner, or the speed at which a vehicle is circulating [5]. Second, there is no additional fine for recidivism.

The central database I will use for defining treatment includes all vehicles that exceeded the legal speed limit under a Speed Camera from August 16 to October 15, 2024. This dataset is comprised of 181,489 distinct vehicles that passed the speed Cameras at the time at which they give speeding tickets, which corresponds to daylight. Since I will only be using vehicles that passed under the Cameras once and had not received a speeding ticket in the previous six months (in order to separate the clean the effect of the previous interaction with the law) a total of 158,703 comprise the full sample used for estimation [6]. This is equivalent to 87% of the plates that exceeded the limit. I will call this sample my Full sample. The robustness check I show that nothing changes if I use vehicles that passed more than once.

I can also access the databases for all existing tickets and crashes occurring in Bogotá [7]. I can see if each plate received a ticket or participated in a crash both before and after the period of study, until March 15 of 2025. Those two will be my main outcome variables but also are important in correcting for specific characteristics of the sample.

[3] I have measured the general impact of the program previously using a staggered intervention design. I find that it reduces crashes, especially the less serious ones (Ferro, 2024)

[4] this is ruling C-321 of 2022 by Corte Constitucional del Colombia, Colombia's Supreme Court

[5] technically there is an early payment, but all vehicles are subject to this, so this does not affect the differences on relative price

[6] Full distribution of the number of observation by vehicle is in this sample is shown on Figure 1. This is a particular problem since vehicles might circulate one time below the real limit and another one above the real limit, thus making it harder to classify them as treated or untreated in terms of the discontinuity. I will do robustness checks on this, but mostly just use vehicles that circulated once

[7] For the crash data I also use the data for the centralized health system that is specifically dedicated to traffic events. This data is under a system called SIRAS, and belongs to the national Minister of Transportation, not the city of Bogotá

I can access the registration record for all vehicles in Bogotá (and some that pay congestion pricing [8]) and match the plate with information on brand, model and year of the vehicle. I do a web scrapping procedure of the most used website for selling used cars in the city to obtain the commercial value of each brand-model-year combination. All data on vehicle prices is taken from tucarro.com, the most used website for selling used vehicles. Vehicles that do not appear on that base are given an imputed value from a regression on year and cubic capacity on price for each class-brand combination.

This leaves me with a total of 67,235 vehicles for which I can find the commercial prices, naturally a subsample but is big enough for the estimation to have statistical power. I will call this sample my Matched sample [9], and use the price of the vehicle as a proxy for wealth. The heterogenous effect of the relative price of the ticket can thus be estimated. Local effects and external validity questions will be analyzed in the Results section.

3 Estimated equation

My estimation derives from a fact that is unknown to drivers, that tickets are not given at the speed posted on the road. On the following I will call the legal limit posted on the road sign (50 km/h) as the posted speed limit, and the limit at which most tickets really given ($50 + \epsilon$ km/h) as the real limit[10].

This discontinuity allows for some exogenous variation that can be exploited. In most of my applications, I estimate the following equation:

$$\text{Result}_i = f(z_i) + \tau D_i + \varepsilon_i \tag{1}$$

Where Result_i is one of my three behaviors of interest for vehicle i: a dummy that takes the value of 1 if the vehicle had at least one instance of excess speed in the six months after receiving a ticket and 0 otherwise [11], a dummy that takes the value of 1 if the vehicle had one crash in the six months after receiving a ticket and 0 otherwise, or the total number of instances of excess speed in the six months after receiving a ticket.

[8] Some vehicles can pay to avoid a circulation restriction that exists in the city (Pico y placa Solidario). All the data I need is registered on these transactions

[9] The decision to register the vehicle in Bogotá is surely related to behavior as the fact that the registry is growing in surrounding municipalities but not in Bogotá. This is particularly caused by the decision of new motorcycle owners to register their vehicle outside Bogotá, even if this is where they plan to use it. Thus, the effect seen on that sample might only extend to other drivers that decide to register their vehicles in Bogotá.

[10] In the sample there is only one Camera that has 30 as the maximum speed. It only accounts for 4,460 observations (1.82%). Speeds are adjusted accordingly. By agreement with the Mobility Office of Bogotá, I will not reveal the specific value of ϵ in order to protect public policy

[11] Since the real limit is unknown to drivers, a future excess speed is the one at above the posted limit, not the real one

$f(z_i)$ is a polynomial of degree 2 (Gelman & Imbens, 2019) for the variable that decides treatment, the speed at which vehicle i is circulating under a Speed Camera. As I said, all vehicles in the sample exceed the posted speed limit, but only some exceed the real limit. This reduces selection and allows me to identify causality.

D_i is my treatment variable, which takes the value of 1 if the vehicle received a speeding ticket and 0 otherwise. The treatment is instrumented via the assignment, since this is a fuzzy design, and the discontinuity at the real speed limit is around an increase of 27 p.p. on the probability of receiving a ticket.

ε_i is an error term.

τ is my coefficient of interest. I will argue that it captures the causal effect of receiving a ticket in subsequent behavior. It is a local effect, since it only captures the effect of a ticket for vehicles around the cutoff, but those are the vast majority of observations in the sample [12]. For all estimations I use the rdrobust package in STATA with the option to ignore mass points, as the variable that assigns treatment is discrete. The bias-corrected estimate will always be shown.

When studying the heterogeneous effect of relative price on behavior I will run regression (1) at each quintile level. This allows for better flexibility of the assignment variable and control for more variables.

3.1 Validation of the Fuzzy Regression Discontinuity Design: Existence of discontinuity:

Figure 1 shows that enforcement does work with the real limit, not the one posted on the street signs. It also shows that the discontinuity is not 0 to 1, so my strategy will be a fuzzy design. The reason behind this jump not being to 1 is mainly the quality of the pictures taken by the cameras. A police officer verifies that the picture was sufficiently clear to transform into a ticket. I will later use the human component of this analysis to verify my estimation. In short, even if the behavior of driving at ($50 + \epsilon$ km/h) is very similar to driving at ($50 + \epsilon + 1$ km/h), and both are above the legal limit, the latter has four times the probability of receiving a ticket than the former.

[12] 80% of observations pass under Speed Cameras inside the bandwidths selected

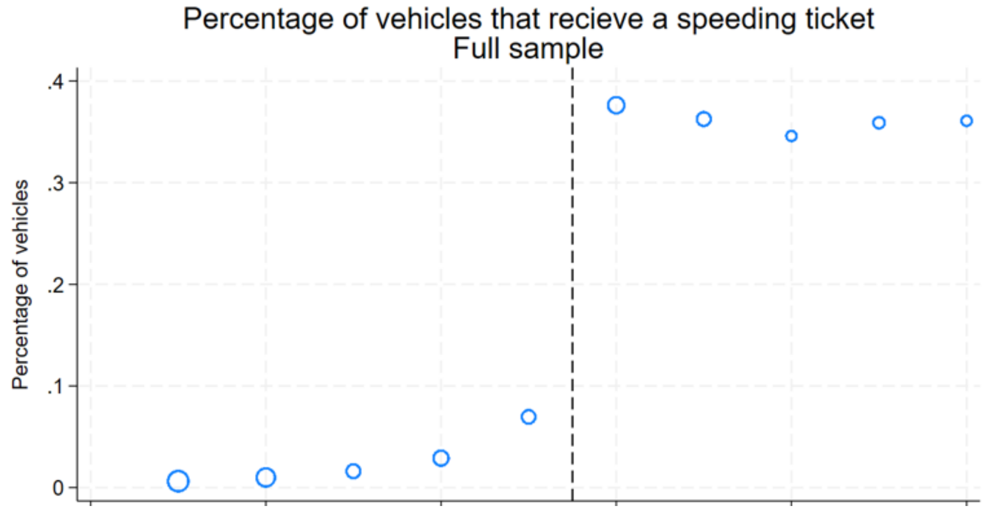

Figure 1: Speed above limit is constructed as: vehicle observed speed-speed limit

It is important to show that this discontinuity also exists in the matched sample and is similar for different values of vehicle prices. Figure 2 plots this discontinuity for all matched vehicles and for each quintile of vehicle price. In fact, the discontinuity is slightly bigger in this case across the board, but particularly for higher quintiles.

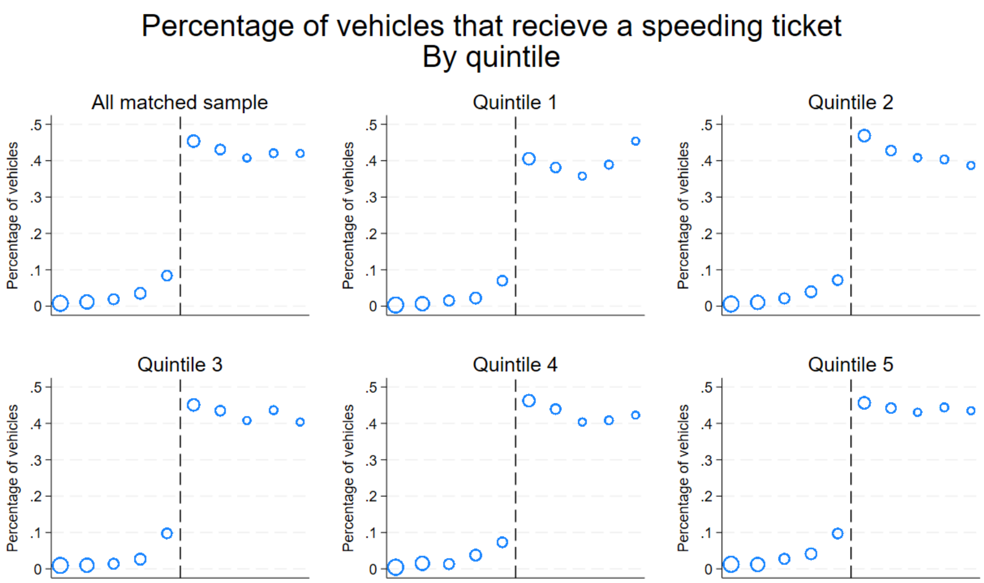

Figure 2: Speed above limit is constructed as: vehicle observed speed-speed limit

A very relevant question for local public policy is whether this deterrent effect of tickets varies by vehicle type. This is caused by motorcycles being a growing percentage of road injuries and fatalities both in Bogotá and in all Colombian cities, as in similar middle income countries. Figure A.2 and Figure A.3 show this for the Full and Matched sample. It can be seen that the discontinuity is smaller for motorcycles, and that this is considerably smaller in the case of the matched sample. This is less a threat for the identification strategy and more of a fact to have in mind when interpreting results.

3.2 No manipulation

Now that I have shown that there is a discontinuity at cutoff, I need to show that there is no manipulation. In short, drivers are not aware of this fact. Since the estimation wants to separate the effect of a possible ticket from any other change, I will only use the vehicles that have not received a ticket in the 90 days before passing under a Camera exceeding the legal speed limit. Frequency is shown in Figure 3 for the Full Sample and Figure 4 for the Matched Sample. In Figure A.4 the same distribution is shown for all vehicles that did receive a ticket in the prior 90 days, which I don't use for different reasons in my main estimation. It seems that even drivers that have had an interaction with the law are not selecting themselves into not receiving a ticket. The same is true for vehicles that speed more than once in the study period. This group, arguably more speed enthusiastic, should bunch more at the real limit. But again, if anything the graph (not shown) shows there is some bunching to the right of the real limit.

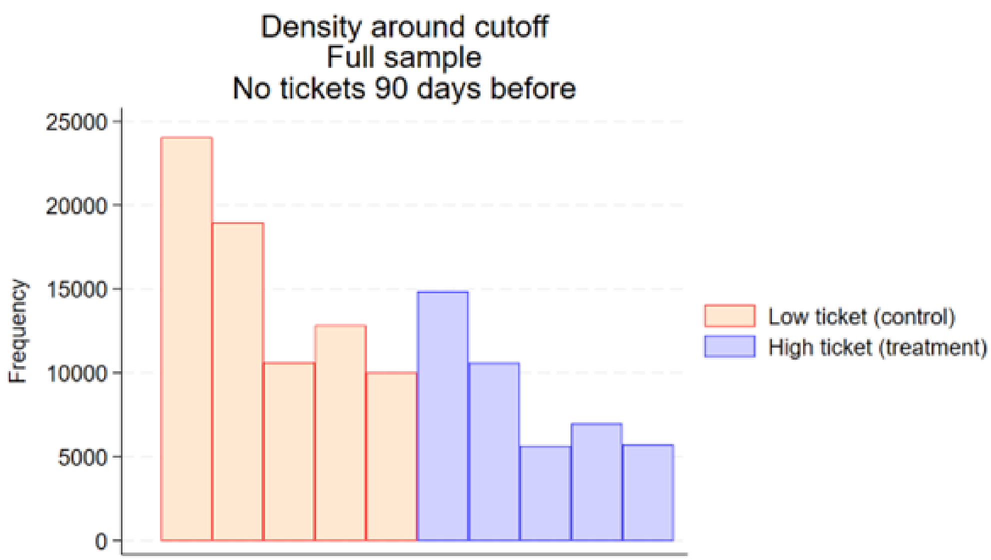

Figure 3: Frequency by condition. Full sample.

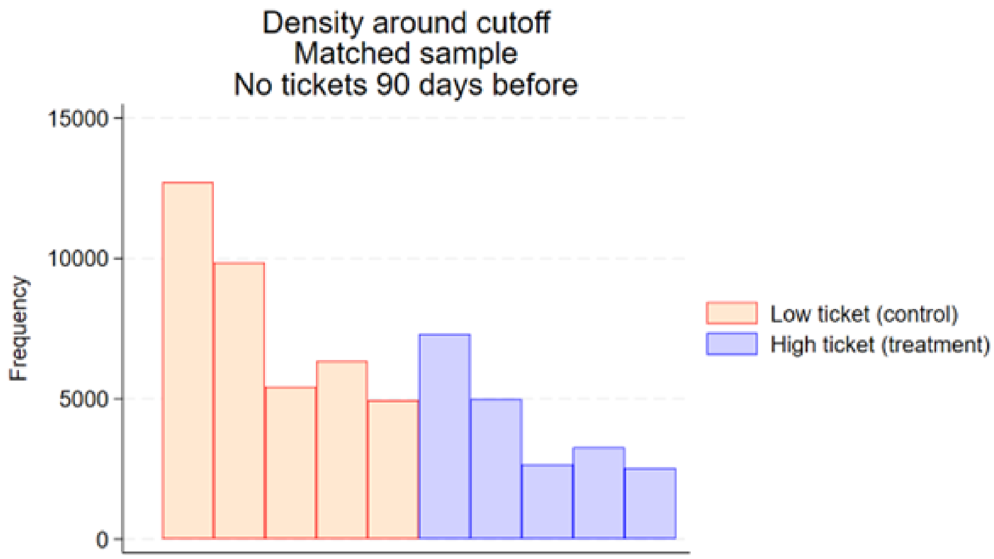

Figure 4: Manipulation test based on discontinuity at the real limit. Sample with polynomial of degree 2. Full Sample.

More formal evidence of this can be given by running the manipulation test developed by (Cattaneo et al., 2018) shown in Figure 5. This is because of the discrete values of the assignment variable that does not allow for a McCrary test to be performed. Figure A.5 performs this test with different decisions regarding the polynomial degree. Figure A.6 shows that the test is also not significant in the matched sample. More importantly, all graphical evidence shows if there is a discontinuity it is to a higher density at the right of the cutoff. This is no evidence manipulation since in principle drivers do not want to get a ticket, so the result of these tests is unlikely to invalidate the exogeneity of the real limit.

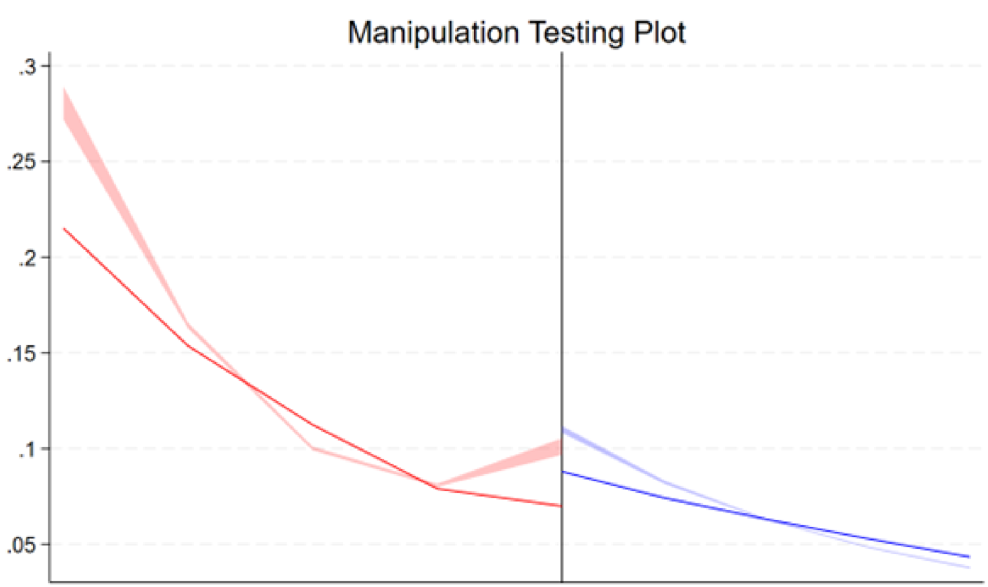

Figure 5: Frequency by condition. Full sample.

Another way to show that drivers are not selecting themselves into not receiving a ticket is to see what happens at the posted limit. One can see in Figure 6 that there is bunching to the left of the posted limit, but to the right of the real limit. This is ratified by Figure 7 in which I use the posted limit as the discontinuity and find that there is the null hypothesis of continuity can be rejected, since vehicles do bunch to the left both at their first observation and with total observations. So vehicles select into treatment at the posted limit, but not at the real one.

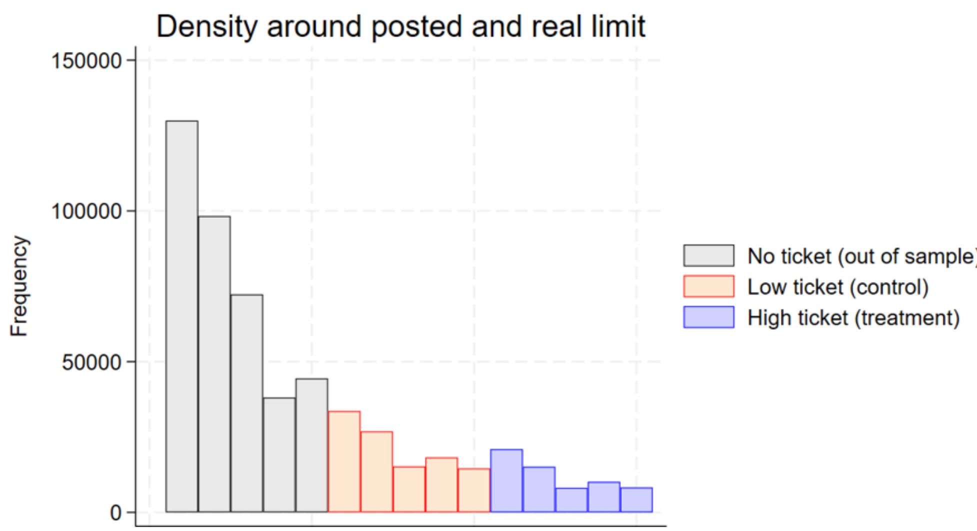

Figure 6: Discontinuity around posted and real limit

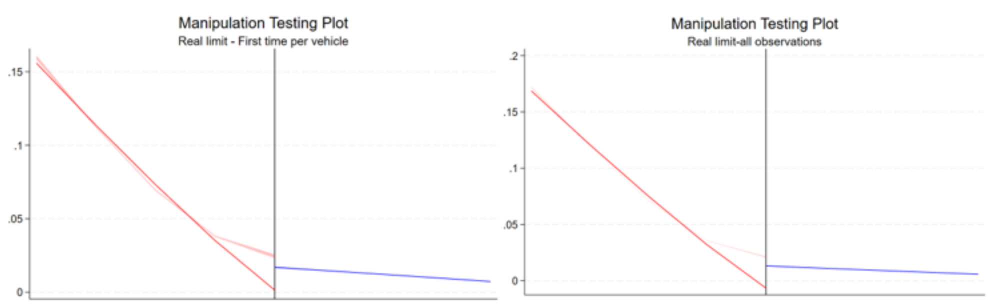

Figure 7: Discontinuity around posted and real limit

In the analysis of the relative price, I will use the Quintiles of the distribution of vehicle price I also need to show that there is no bunching at the quintile level. **??** shows this for each quintile, and Figure A.7 of the Appendix shows the same data for vehicles with a previous ticket 90 days before. One can ratify that even drivers with previous ticket are not learning what the real limit is. In fact, even taxis and other public service vehicles do not show bunching to the left of the real limit (figure not shown).

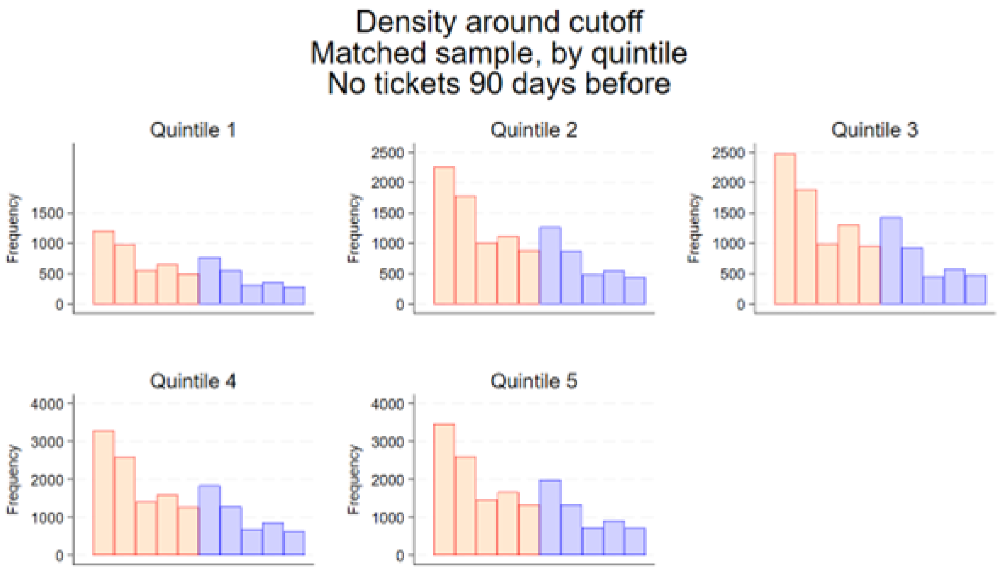

Figure 8: Frequency by condition and quintile. Matched sample.

3.3 Smoothness on observables

3.3.1 Full sample

Another element I need to evaluate the validity of the fuzzy regression discontinuity design is to measure the smoothness of observable variables of the vehicles involved. I will do this first on the full sample, and then on the matched sample, in which naturally I have more knowledge of the vehicle characteristics. For all estimations I allow for optimal and different bandwidth below and above the cutoff.

The first variable I explore is the hour of the day at which the ticket is received, a possible indications that drivers are different in the treatment and control groups. I can even compare if there is discontinuity in the provability of being in rush hour traffic, finding a slight difference. The same selection can be analyzed with day of the week as a continuous variable or a dummy if the ticket was given during the weekend. Since in Colombia, plates for motorcycles have a different configuration than those of other vehicles [13], I can observe the type of vehicle involved. I use a dummy that takes the value of 1 if the vehicle involved is a motorcycle. This variable is in fact discontinuous, and magnitude is considerable. Selection is a big concern when analyzing fines and is the reason I estimate an RD in the first place. In any case, it is important to observe that treated and not treated do not differ in their road usage. I compare the number of times each vehicle passes under a Camera not speeding, before the first speeding infraction. This is a good proxy for their road usage, and one that is discontinuous at the real limit, but with a smaller magnitude than the motorcycle dummy.

[13] Motorcycle plates end in a letter, whether all other vehicle's plates end in a number. As will be seen later, for the matched sample I have much more knowledge about the vehicle.

In the following I will add type of vehicle and previous observations as covariates that affect the treatment status. All this is shown in Table 9.

Smoothness on observables. Full sample

	(1) Hour of the day	(2) Rush hour dummy	(3) Day of the week	(4) Weekend dummy	(5) Motorcycle dummy	(6) Observations per day before
Bias-corrected	0.188	-0.0579*	0.0146	-0.0345	-0.0867***	-0.569**
	(0.248)	(0.0335)	(0.148)	(0.0338)	(0.0336)	(0.274)
Mean of not ticketed	11.80	0.350	4.156	0.350	0.341	3.682
N	136533	136533	136533	136533	136533	136533
N left of cutoff	52468	52468	52468	52468	52468	52468
N right of cutoff	57605	57605	57605	57605	57605	57605
BW left	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Smoothness on observables. Full sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Rush hour is a dummy that takes the value of 1 if the vehicle passed under the camera at rush hours (6,7 am or 4,5,6 pm) and 0 otherwise. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess

To better understand the composition of my sample I can also do a smoothness analysis on the real speed limit, not as a formal test but to understand what type of vehicles are selecting themselves into treatment [14]. This analysis shows that, using the first time seen under a speed camera, speeders are most common early on the week, and on weekdays. Also, unlike at the real limit the coefficient for motorcycles is positive, suggesting that drivers motorcycles are more prone to speed. Additionally the coefficient for future observations is negative, which can be interpreted as vehicles that decide to speed are less frequent road users [15]. All this points to the sample of speeders being different, but not to different than non-speeders.

[14] I can also add that speeders are much more crash prone that non-speeders. Using all observations in this period, speeders are twice more likely than non speeders to ever be involved in a crash.

[15] One can do this same analysis not with the first observation by plate, but by observations. Total observations around the posted limit are more than 20 million, and none of the variables are discontinuous. This is hard to interpret as proof that vehicles that speed are similar than those that do not, but differences are very subtle

3.3.2 Matched sample

In this case the total number of available observations is 67,235 [16], around 40% of the Full Sample. Just like with the full sample, most of the variables are smooth at the cutoff. It is also the case that vehicles in the sample tend to come not from the rush hour hours (and there is significance). There are also less motorcycles at the right of the cutoff, but one should note the significant difference in percentages between the two samples. Whereas in the Full Sample 34% of vehicles are motorcycles, this percentage is only 7% in the matched sample. Note also that the Matched Sample has a higher number of observations before the first excess, meaning that these vehicles are more frequent users of the road than the general sample. Both facts reinforce my argument about the Matched sample not being a representative part of the Full Sample.

Note in Table 10 that there is no significant difference in terms of observations before the speed excess and price. Both facts are also central to my estimation later on. In this case I can also evaluate the type of service which the vehicle is assigned to through a dummy that takes the value of 1 if a vehicle is used for public service (mainly taxis and similar services). I can also say a little more about the vehicle type, besides not being a motorcycle. In this case I construct a dummy that takes the value of 1 if a vehicle is an SUV like vehicle. Both of these additional variables are smooth at the cutoff.

Smoothness on observables. Matched sample

	(1) Hour of day	(2) Rush hour dummy	(3) Day of week	(4) Weekend dummy	(5) Motorcycle dummy	(6) Observations per day before	(7) Price of vehicle	(8) Service type	(9) Vehicle dummy
Bias-corrected	-0.290	-0.0904**	0.196	-0.0193	-0.0582***	-0.232	-7.594*	-0.0203	-0.0633
	(0.287)	(0.0388)	(0.179)	(0.0406)	(0.0211)	(0.364)	(4.428)	(0.0294)	(0.0388)
Mean of not ticketed	11.95	0.336	4.320	0.402	0.0702	4.747	56.86	0.153	0.328
N	67235	67235	67235	67235	67235	67235	67235	67235	67235
N left of cutoff	26547	26547	26547	26547	26547	26547	26547	26547	26547
N right of cutoff	26911	26911	26911	26911	26911	26911	26911	26911	26911
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Smoothness on observables. Matched sample. Hour is a continuous variable with the time in which the vehicle passed under the camera. Rush hour is a dummy that takes the value of 1 if the vehicle passed under the camera at rush hours (6,7 am or 4,5,6 pm) and 0 otherwise. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess

[16] As I explained before, the vast majority of those (62,288 or 92% of the matched sample) correspond to vehicles that are registered in Bogotá. The rest are registered in other municipalities but pay an exemption for the congestion restriction that exists in the city that allow local authority access to the vehicle data

Also central to my estimation on the heterogeneous effect of relative price is verifying the smoothness at each quintile level. This is shown on Table A.8 to Table A.8. In general, there is smoothness, but it increases as the quintile level goes up. Since the behavioral explanation of my findings depends heavily on what happens at quintile 5 this is reassuring for my estimation strategy. In any case, as I do in my Full sample all the observables that are significant will be used as covariables for treatment assignment.

4 General results

In this section I show all the results of the estimations. In section 4.1 I presents all results on direct deterrence – speeding. In Section 4.2 I presents results on indirect deterrence – crashes. Chapter 5 is dedicated to the relative price analysis.

4.1 Direct deterrence – Speeding

The most basic deterrent effect of a speeding ticket is to prevent speeding itself. Note that the assumption that drivers do not know the real speed limit implies that reoffending is measured by exceeding the posted limit. Since the risk of injury and death grows with the level of speeding it is also important to measure the deterrent effect on the most serious infractions.

4.1.1 Full sample

Table 11 presents the results for equation (1) with different dependent variables related to speeding for a period of six month after a vehicle is observed speeding under a Camera. Moderate Speeding relates to the fact that a vehicle was observed at a speed between 50 km/h and 60 km/h at least once. Major speeding (Speeding¿10 in the table) refers to the fact that a vehicle was observed at a speed higher than 60 km/h at least once. Odd columns refer to a dummy that measures if the vehicle ever had the behavior (extensive margin), while even columns measure the total number of times each vehicle had the behavior (intensive margin). Most, but far from all, vehicles commit a speeding infraction only once.

The results point at ticket having a causal effect of reducing all types of speeding behavior. This reduction is significant in magnitude, both at the extensive a (77% reduction in the change of reoffending) and intensive margins. Very important is the fact that the result stands for major speeding, even if its probability is a quarter of the probability of moderate speeding. I can even see (results not reported) that a fine has a negative effect on average speed in the future, pointing at drivers not only wanting to avoid fines or bunching at speed limits.

Effect of receiving a ticket

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.175***	-0.295***	-0.0651***	-0.0744***
	(0.0294)	(0.0639)	(0.0164)	(0.0250)
Mean of not ticketed	0.226	0.370	0.0555	0.0716
N	136533	136533	136533	136533
N Left of cutoff	52468	52468	52468	52468
N Right of cutoff	57605	57605	57605	57605
BW left	4.5	4.5	4.5	4.5
BW right	9.5	9.5	9.5	9.5

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11: Direct deterrence. Full sample. Column 1 is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 is a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. Column 4 is a continuous variable, the total of times a vehicle was captured in Major Speeding

A central question for policy makers is the effectiveness of speeding tickets and other measures for speed control in the face of the growing number of motorcycles in middle income countries, and the risk they represent [17]. As I explained before, the existing data only allows me to build a dummy that says if a vehicle is a motorcycle (=1) or not (=0). As is shown in Figure A.2 of the Appendix, even if the discontinuity is smaller for motorcycles, it is still present for both values of my dummy variable. Table A.13 and Table A.14 show that deterrent effect is only present for not-motorcycles, which is in line with my behavioral mechanisms since motorcycle drivers are a group known to consider traffic enforcement of speed limits an unfair action by the government.

[17] In Bogotá, motorcycle users were present in 45.5% of crashes with a death in 2015. This number has steadily grown to 68.5% in 2024

4.1.2 Matched sample

I can also do this analysis in the matched sample. As Table 12 shows the results are essentially the same, a general effect of reduction in all types of speeding. Note also that the result for Major Speeding is slightly smaller in magnitude and less robust than in the general sample.

Effect of receiving a ticket

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.186***	-0.254***	-0.0531***	-0.0554**
	(0.0361)	(0.0729)	(0.0191)	(0.0265)
Mean of not ticketed	0.228	0.337	0.0565	0.0687
N	67235	67235	67235	67235
N Left of cutoff	26547	26547	26547	26547
N Right of cutoff	26911	26911	26911	26911
BW left	4.5	4.5	4.5	4.5
BW right	9.5	9.5	9.5	9.5

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12: Direct deterrence. Matched sample. Column 1 is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 is a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. Column 4 is a continuous variable, the total of times a vehicle was captured in Major Speeding

4.2 Indirect deterrence – crashes

As I argued in the introduction, speeding tickets have an indirect deterrent purpose explained by the fact that speeding both makes crashes more probable and more severe. Measuring crashes in the whole city is a good proxy for whether drivers decide to speed less in the vast majority of locations where they know they will not be ticketed for doing so. This is the real test for the behavioral change that tickets are designed for.

Table 13 presents the results of equation (1) when the dependent variable measures crashes. Neither a dummy for having a crash (column 1), nor the continuous variable for total of crashes (column 2), nor the dummy for crashes with at least one fatality [18] (column 3) are significant. Nor is it useful to separate the sample into cars and motorcycles. The effect on the matched sample is equally null in all columns, so I do not report it.

Effect of receiving a ticket

	(1)	(2)	(3)	(4)	(5)
	Any crash dummy	Total crashes continuous	Deadly crash dummy	Cars dummy	Motorcycles dummy
Bias-corrected	-0.00143	-0.00210	0.00441	-0.00337	0.000703
	(0.0131)	(0.0166)	(0.00359)	(0.0100)	(0.0328)
Mean of not ticketed	0.0363	0.0420	0.00258	0.01317	0.08106
N	136533	136533	136533	89037	47496
N Left of cutoff	52468	52468	52468	35012	17456
N Right of cutoff	57605	57605	57605	35827	21778
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13: Indirect deterrence. Full sample. Column 1 (Any crash) is a dummy that takes the value of 1 if the vehicle had at least one crash in the following months, and 0 otherwise. Column 2 (Total Crashes) is a continuous variable; the total of times a vehicle had a crash. Column 3 (Deadly Crash) is a dummy that takes the value of 1 if the vehicle had at least one crash with a fatality or Triage I attention in the following months, and 0 otherwise. Column 4 (Cars) is the same as Column 1 restricting the sample to cars. Column 5 (Motorcycles) does the same for motorcycles.

A possible objection to these results could be the fact that the estimation is underpowered. I should not that the same number of observations produced a very credible result for speeding. I should also point out that the number of crashes in the sample is small, but not negligible. There are 7,162 (108 of them deadly) crashes in my estimation sample. These are around 6.2% of the total 116,088 total crashes that occurred in the period of study. In addition, I would say that 3.6% of vehicles in the control sample do have a crash, so it is a rare but not extremely rare phenomena. I would argue that the effect is too small to be detected, where it big my estimation would detect it.

[18] I also include here those crashes that required immediate medical attention because of imminent risk of death, Triage I, according to the Triage categories defined by the Colombian Minister of Health. 8% of the crashes that required medical attention are considered Triage I

5 Relative price

The central question of this paper is whether the deterrent effect of fines is independent of their relative price. Many countries have fines that are relative to income or wealth, mainly out of justice considerations. I am thinking mainly about effectiveness for preventing unwanted behavior. Table 14 reproduces the main estimations on speeding and crashes separating vehicles that are below and above the median of the commercial price of the vehicle as proxy for the owner's wealth. Coefficients show that vehicles below the median the ticket has no direct deterrence effect and an unintended result for indirect deterrence, while for vehicles above the median the ticket has the desired effect both in direct and indirect deterrence. One could argue that the desired effects of a speeding fine are concentrated on the most wealthy, while fines have undesired effects on the less wealthy. Note that the result on indirect deterrence is particularly worrisome since vehicles below the median have a higher average rate of crashes, so fines make worse the situations for the ones where deterrence is needed most.

Table A.15 reproduces the estimation below and above the median with the most costly results as dependent variables. There is no change on the effect of direct deterrence only being present above the median, while there is a backfiring effect for both groups when talking about deadly crashes. Both the magnitude of the coefficient and the base rate is much higher for vehicles below the median, so even if there is backfiring in both groups once again the policy is making worse the conditions of those groups where deterrence is more needed.

Effect of receiving a ticket

	(1)	(2)	(3)	(4)
	Below median	Above median	Below median	Above median
	Minor	Minor	Any	Any
	Speeding	Speeding	crash	crash
Bias-corrected	-0.0509	-0.267***	0.0381**	-0.0346**
	(0.0552)	(0.0470)	(0.0180)	(0.0141)
Mean of not ticketed	0.218	0.268	0.0207	0.0154
N	24793	42442	24793	42442
Price in dollars	5379	19462	5379	19462
N left of cutoff	9837	16710	9837	16710
N right of cutoff	9937	16974	9937	16974
BW left	4.500	4.500	4.500	4.500
BW left	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Direct and indirect deterrence. Matched sample. Columns 1 and 2 (speeding) are a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Columns 3 and 4 are a dummy that takes the value of 1 if the vehicle had at least one crash, and 0 otherwise.

A natural question if the result holds when using alternatives distributions of the relative price variable. I can separate the relative price variable by quintiles. Table 15 reproduces equation (1) with the dependent variable of a dummy of speeding (column 1 of Table 11). One can see that the effect is concentrated on the higher quintiles of the distribution. Both magnitude and significance are lower for the lower quintiles, in fact the effect is increasing on the quintile of vehicle price and significant above Quintile 2 [19]. But Table A.17 also shows a complex picture when using Major Speeding. In this case, the effect is not higher in the higher quintiles, but on Quintile 2. I will come back to this puzzle when discussing the effect of speeding tickets in other traffic infractions.

[19]As Table A.16 shows this result is robust to using the continuous variable of the total times a vehicle is captured under Minor Speeding, with the most negative coefficient also found in the highest quintile

	(1)	(2)	(3)	(4)	(5)
	Quintile 1 Minor Speeding	Quintile 2 Minor Speeding	Quintile 3 Minor Speeding	Quintile 4 Minor Speeding	Quintile 5 Minor Speeding
Bias-corrected	-0.0649	-0.0815	-0.195**	-0.213***	-0.213**
	(0.116)	(0.0742)	(0.0879)	(0.0670)	(0.0750)
Mean of not ticketed	0.170	0.240	0.244	0.263	0.278
N	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Rift of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Effect of receiving a ticket

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 15: Direct deterrence by quintile of vehicle price. Matched sample Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one speeding between 50 and 60km/h, and 0 otherwise. For each quintile, variables that are not smooth at cutoff are used as covariates

Table 16 does the same quintile estimation for the dummy of ever being in a crash. The desired effect is increasing in the vehicle quintile but less clearly. Quintile 5 is the only significant and negative result for crashes. But unlike the result on direct deterrence, there is at least some citizens that move in the undesired direction, those on Quintile 2 have a positive coefficient. Note that the rates of crashes are different than those for speeding. While speeding is more common on higher quintiles, crashes are concentrated on the lower ones. Thus, while speeding tickets are well targeted at reducing speeding, they are poorly targeted at preventing injury and fatalities. Table A.18 shows a much more nuanced finding when talking about deadly crashes. In this case coefficient is positive in most quintiles but is of a higher magnitude at quintile 1 than any other, in fact it still decreasing on quintile level. I would advocate for not over interpreting results on deadly crashes, since they in fact are extremely rare.

Effect of receiving a ticket

	(1)	(2)	(3)	(4)	(5)
	Quintile 1	Quintile 2	Quintile 3	Quintile 4	Quintile 5
	Any crash	Any crash	Any crash	Any crash	Any crash
Bias-corrected	0.0307	0.0503***	0.0335	-0.0247	-0.0661***
	(0.0612)	(0.0164)	(0.0223)	(0.0193)	(0.0239)
Mean of not ticketed	0.0443	0.0112	0.0113	0.0142	0.0177
N	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Rift of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16: Indirect deterrence by quintile of vehicle price. Matched sample . Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one crash, and 0 otherwise. For each quintile, variables that are not smooth at cutoff are used as covariates.

All this section points in the same direction, relative price does matter and in the case of Bogotá it seems to suggest that tickets are more effective not when they are more, but less expensive. A natural question for these results is how much they are the effect of taxis and other public service vehicles. I can take them out of the sample and all results (not reported) are in line with the story told in this chapter.

6 Robustness checks

In this section I introduce some further evidence that shows my results are credible. First, I present alternative estimations to show consistency. Second, I use an alternative method of estimating my results: an IV-judge type of estimation using data on the police agent in charge of verifying each speeding ticket.

6.1 Robustness to some decisions

Figure 17 shows a comparison between the coefficient found on column 1 of Table 11 and the coefficient found with different decisions. In particular, whether to include vehicles that had a previous fine [20], to use vehicles that pass more than once under a Camera and to have a bigger time for selecting my sample.

The most important choice is to include vehicles that have more than one excess in my sample period. There are at least two alternatives for including these vehicles. Since the problem is really how to assign treatment, my choice is to consider vehicles that passed always below or above the real limit and use their average speed and first observation as treatment time (Repeated observations, one side on Figure 17). I find this better to the convention of using the first time the vehicle exceeds as its treatment assignment. Since there is a delay between speeding and the moment in which a vehicle is notified of the infraction, a vehicle can be treated and not know it for many days (Repeated observations, first on Figure 17). The coefficient is very stable across all decisions. In fact, including previous observations only increases the magnitude of the effect.

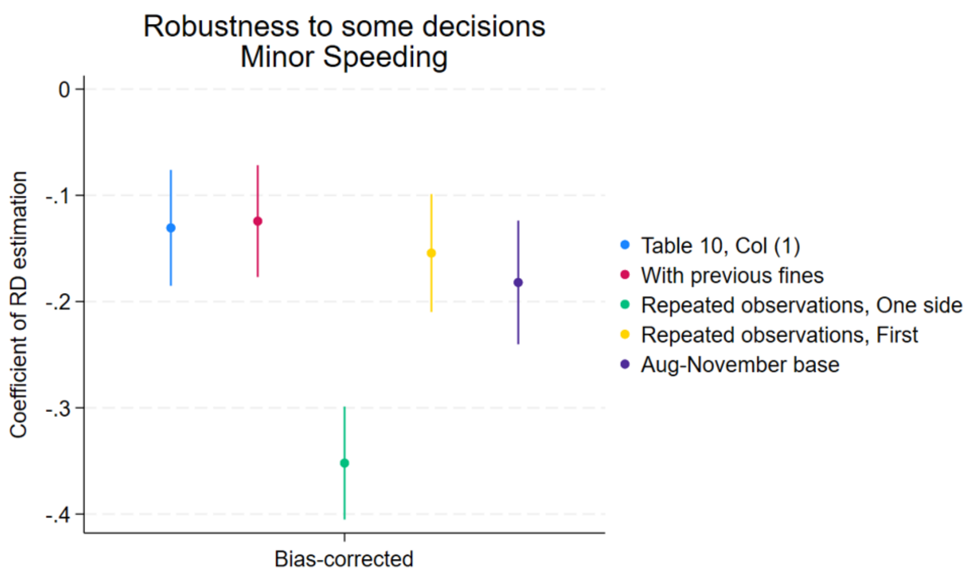

Figure 17: "Use previous fines" uses the same estimation, this time including vehicles that have had a ticket in the previous six months. "Repeated observations, One Side" shows the coefficient using Vehicles that passed more than once and always above or below the cutoff. "Repeated observations, First" shows the coefficient using Vehicles that passed more than once using the first observations as treatment assignment. "Aug-November" base makes the database bigger, by including all vehicles from August 16 to November 15, 2024.

Figure A.18 and Figure A.19 present the same calculations for other dependent variables: Major Speeding and Any crash. The coefficients are very stable to most of the decisions I

[20] As I argued, I want t to separate the effect of this fine from all previous fines. Including those vehicles that had a fine in the previous six months increases the sample to 145145 (from 136553), a 6.2% increase.

make in the general case, but a longer period of study changes some of the results. I should point that smoothness of the variables is less strong on those bigger samples, the reason I do not use them in the first place.

6.2 An IV-judge using police officer manual validation

According to Colombian law all speeding tickets must be subject to human verification. This has to do mainly with the fact that the technology for taking pictures of vehicles while speeding is not perfect, and some pictures are blurry. This task is done by trained police officers that must evaluate whether a specific picture should transform into a ticket. The pictures assigned to each policemen are random, and they have no way to refuse their verification.

I have data for the identification of all pictures of vehicles that were circulating above the real speed limit ($50 + \epsilon$ km/h) during the months of October and November. I also can see which agent was in charge of reviewing each picture. As Figure 18 shows, there is considerable heterogeneity in the percentage of pictures that each agent transforms into a ticket.

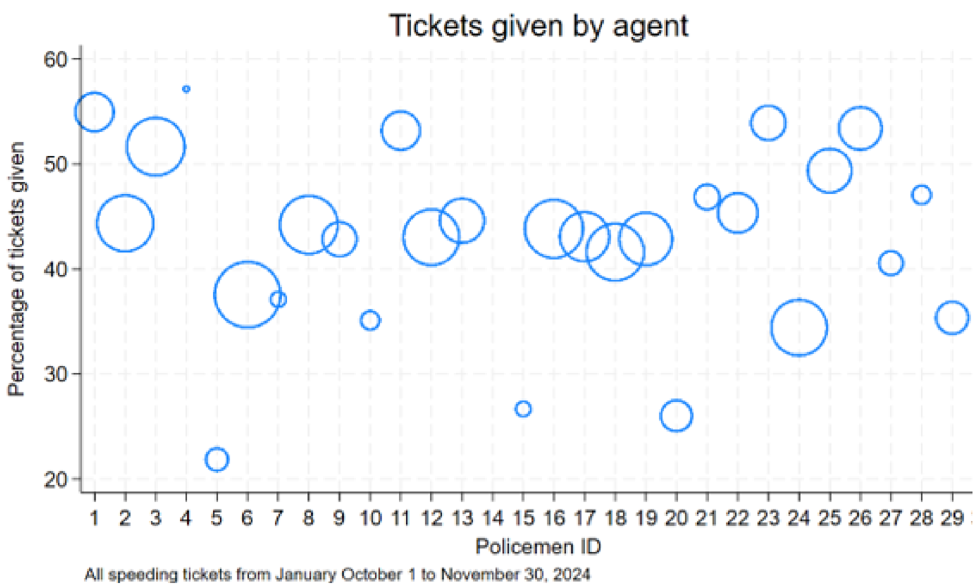

Figure 18: Policeman ID is a number to identify each police officer. The size of the circle is proportional to the number of pictures reviewed by each agent.

Figure 18 indicates that there might be some randomness associated to the Policemen in charge. For this to transform into a credible instrumental variable one would want to verify that these different rates are not related to any observable characteristic of the ticket, or the vehicle involved. Figure 19 shows that the strictness is stable for all Policemen across at least two meaningful variables, whether a vehicle is a motorcycle and those vehicles that drive under Major speeding, as previously defined.

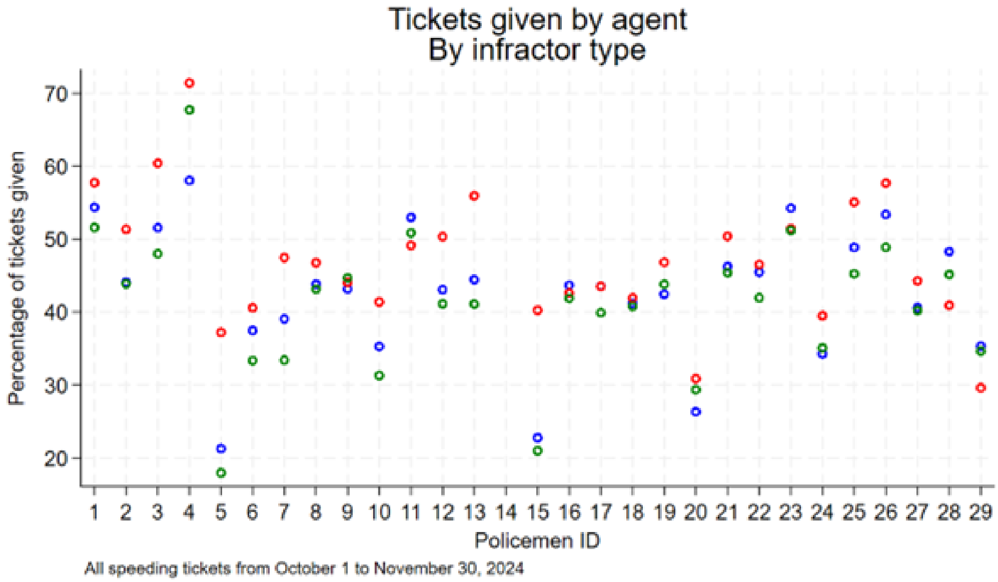

Figure 19: Policeman ID is a number to identify each police officer. Blue dots correspond to all pictures, red dots to pictures of motorcycles and green to vehicles under major speeding (40% of this subsample).

Both previous Figures give evidence to use the random assignment of the Policemen in charge of reviewing each picture as an instrument of the ticket. I will then capture the part of receiving a ticket that is due to the randomness of the assigned policemen. I also must emphasize that this is a different sample, consisting mostly of drivers that decide to exceed speeding limits seriously. I should also note that in this case I am not finding a local effect, but the general effect on those that decide to exceed the limit by a significant amount of kilometers.

Table 20 shows the result of an estimation with the same dependent variables as Table 8. As one can clearly see, the significance and sign of all the estimation is similar, but magnitudes are considerably smaller. While the effect of a speeding ticket was a 77% reduction in the change of reoffending, in this case the effect is only 25%. This surely is caused by these drivers deciding to disobey speeding limits by a bigger amount than the other sample. These drivers like speeding more, so are less likely to slow down after being told to do so.

Effect of agent reviewing ticket

	(1)	(2)	(3)	(4)
	Speeding dummy	Speeding continuous	Speeding >10 dummy	Speeding >10 continuous
Ticket	-0.0416***	-0.0840***	-0.0201**	-0.0257**
	(0.0158)	(0.0279)	(0.00897)	(0.0127)
Mean of not ticketed	0.164	0.241	0.0470	0.0604
N	132668	132668	132668	132668
First stage Wald-F	82.56	82.56	82.56	82.56

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 20: Direct deterrence. Alternative sample.. Column 1 is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 is a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. Column 4 is a continuous variable, the total of times a vehicle was captured in Major Speeding

The results Table 20 are in line with the main finding of this paper. One could also use this sample to verify the second finding: effects being explained only by cars, not motorcycles. Table A.20 and Table A.21 do this. Note that in this case the effect is more precisely estimated for cars than motorcycles, but there is some for the latter. Once again, this does not cast doubt on previous findings but shows that major speeders are more similar between them, independent of the type of vehicle they drive.

One could use this estimation strategy to verify the other two findings. As Table 21 shows there is also a null effect on all dependent variables related to the chance of being involved in a crash. In this case the coefficients are positive in sign, but equally not significant.

Effect of agent reviewing ticket

	(1) Any crash dummy	(2) Total crashes continuous	(3) Deadly crash dummy	(4) Cars dummy	(5) Motorcycles dummy
Ticket	0.00484	0.00837	-0.0000405	0.00118	0.00267
	(0.00664)	(0.00816)	(0.00182)	(0.00463)	(0.0182)
Mean of not ticketed	0.0246	0.0280	0.00194	0.0103	0.0630
N	132668	132668	132668	94536	38132
First stage Wald-F	82.56	82.56	82.56	65.40	28.93

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 21: Indirect deterrence. Alternative sample.. Column 1 (Any crash) is a dummy that takes the value of 1 if the vehicle had at least one crash in the following months, and 0 otherwise. Column 2 (Total Crashes) is a continuous variable; the total of times a vehicle had a crash. Column 3 (Deadly Crash) is a dummy that takes the value of 1 if the vehicle had at least one crash with a fatality or Triage I attention in the following months, and 0 otherwise. Column 4 (Cars) is the same as Column 1 restricting the sample to cars. Column 5 (Motorcycles) does the same for motorcycles. This table is comparable to Table 13.

The final finding one could verify has to do with the heterogeneous effect by vehicle price. As Table 22 shows, the effect of a ticket on speeding behavior is in this case also concentrated on the higher quintiles. Even more, like in previous cases for quintile 2 there is a positive coefficient, which is slightly not significant at conventional levels. Table 21 of the Appendix does the same with any crash as dependent variable, where results are more blurry, in part because of the small relevance of the instrument or the difference of the sample composition.

Effect of agent reviewing ticket

	(1)	(2)	(3)	(4)	(5)
	Any crash	Total crashes	Deadly crash	Cars	Motorcycles
	dummy	continuous	dummy	dummy	dummy
Ticket	0.00484	0.00837	-0.0000405	0.00118	0.00267
	(0.00664)	(0.00816)	(0.00182)	(0.00463)	(0.0182)
Mean of not ticketed	0.0246	0.0280	0.00194	0.0103	0.0630
N	132668	132668	132668	94536	38132
First stage Wald-F	82.56	82.56	82.56	65.40	28.93

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 22: Direct deterrence by price quintile. Alternative sample.. Dependent variable is a dummy that takes the value of 1 Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one speeding, and 0 otherwise.

In summary, although the sample for the previous sections is different from this one, nor are the estimation strategies capturing the same causal effect, one can safely say that these results give more credibility to the causal effects found. All evidence points in the same direction of tickets having a deterrent effect, and this effect being bigger for higher quintiles of vehicle prices.

7 Wake-up call or information? An exploration on mechanisms.

Speeding tickets are meant to prevent the most serious consequences of traffic crashes. In fact, speeding Cameras in Bogotá where originally called "Camaras Salvavidas" (Lifesaving cameras). As we have seen, in practice Cameras seem very effective at reducing speeding itself but show little effect on reducing crashes. In this section I use other behaviors to try to understand how much of the effect of traffic tickets corresponds to changes in the driver's behavior and how local, both in geographical and chronological sense, the effect is.

The first question I want to explore is wether the results are caused by a reduction in the frequency of driving. This is not good or bad in itself in terms of road safety, but it might have significant costs at the personal level. I use as a dependent variable the total number of times a vehicle passes under a Speed Camera after the speeding infraction. Column 1 of Table 23 shows that speeding tickets seem not to have an effect on road usage, and that this mechanism is surely not causing the results seen before.

A second question I can address is how local the effect is. For this purpose I can use the location of each Camera. Note that due to the nature of traffic management in the city of Bogotá all Cameras are not active during the six months following an speeding infraction. Instead of measuring each Camera I use the division of Bogotá into 20 Localidades (something close to Boroughs). In Columns 2 and 3 of Table 23 I separate further speeding into close or far from the Camera at which a vehicle was captured speeding. I consider close any camera that is inside two levels of proximity, and far those occurring outside this range. Result is robust to using just one level of proximity.

A third question, and the most interesting one in behavioral terms, is whether in general speeding fines are causing a change in driving behavior or only the need for being generally more aware of the locations and speed limit of Speed Cameras. This question can be addressed by looking at other behaviors on the road that are also punished by road authorities. This is complicated by the fact that speeding is by far the most common infraction in the period of study, even if this period coincides with a city wide campaign against parking in prohibited spaces. Column 4 and 5 of Table 23 explore both the extensive and intensive margin of this dimension. It is reassuring to see that this happens, since it suggest a broad behavioral change.

Table A.23 examines how the result seen on Column 4 changes by quintile. There is an interesting finding since, contrary to all other effects, the most desired outcome (a big reduction in other tickets) is concentrated on quintiles 2 and 3. A tentative explanation for this finding has to do with the fact that Speeding tickets have no interaction with a policemen, nor the risk of confiscation of the vehicle, whereas other tickets have this risk. So a speeding ticket might be a sign that the authority is present, even if it is incapable of collecting the money from speeding fines. This hypothesis of the risk of engaging in person with the police goes in line with the complex effect of tickets on Major Speeding, another behavior that might alert human policemen. Al this points to other tickets being an alternative form of enforcement that drivers want to avoid and to show that while not effective at the most desired outcome, the global effect of speeding tickets has to consider a more general picture of behavior and sense of authority. Note that in Bogotá is illegal for policemen to look at past behavior of a vehicle while giving a fine, so it is unlikely that the reduction comes from police discretion.

Behavioral mechanisms Full sample

	(1) Future observations	(2) Minor Speeding nearby	(3) Minor Speeding far	(4) Other tickets dummy	(5) Other tickets continuous
Bias-corrected	2.359	-0.105***	-0.0691***	-0.0867***	-0.125***
	(1.963)	(0.0267)	(0.0156)	(0.0228)	(0.0387)
Mean of not ticketed	21.013	0.176	0.0504	0.117	0.160
N	136533	136533	136533	136533	136533
N Left of cutoff	52468	52468	52468	52468	52468
N Right of cutoff	57605	57605	57605	57605	57605
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 23: Behavioral change. Full Sample.. Column 1 is a continuous variable that measures the total number of observations after the first speeding of a vehicle. Column 2 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h) inside a range of two proximate Localidades and 0 otherwise. Column 2 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h) outside a range of two proximate Localidades and 0 otherwise. Column 4 is a dummy variable that takes a value of 1 if the vehicle ever had another ticket for any cause after the first speeding instance, and 0 otherwise. Column 5 is a continuous variable that measures the total number of ticket for any cause after the first speeding instance.

There is also a final question about persistence of the effect of receiving a ticket in speeding. Table 24 shows that the effect is not particularly concentrated on the first months, but it seems to fade out after six months. One can say that it is reassuring that there is a significant effect after five months. Table 22 of the Appendix does the same for Major speeding. Once again, in this case the story is much more blurry. But with some confidence one can say that the effect is present for more than three months after the day of the ticket.

Effect of receiving a ticket. By month

	(1) Minor Speeding any time	(2) Minor Speeding Month 1	(3) Minor Speeding Month 2	(4) Minor Speeding Month 3	(5) Minor Speeding Month 4	(6) Minor Speeding Month 5	(7) Minor Speeding Month 6
Bias-corrected	-0.175***	-0.0216*	-0.0811***	-0.0139	-0.0481***	-0.0466***	0.0187*
	(0.0294)	(0.0127)	(0.0158)	(0.0153)	(0.0153)	(0.0139)	(0.00971)
Mean of not ticketed	0.222	0.0352	0.0539	0.0488	0.0481	0.0382	0.0194
N	136533	136533	136533	136533	136533	136533	136533
N left of cutoff	52468	52468	52468	52468	52468	52468	52468
N right of cutoff	57605	57605	57605	57605	57605	57605	57605
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 24: Behavioral change. Full Sample.. Column 1 (speeding) is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h) any time after the first speeding instance and 0 otherwise. Columns 2 to 7 measure the same variable by month.

8 Discussion and possible policy recommendations

Gary Becker wrote that "Fines are preferable to imprisonment and other types of punishment because they are more efficient" (Becker, 1992). In his analysis "the social cost of fines is about zero", and the optimal amount of a fine "depend[s] only on the marginal harm and cost and not at all on the economic positions of offenders"(Becker, 1974). To be fair, Becker was surely also thinking of more serious crimes that a speed of $50 + \epsilon$ km/h, but those strong statements are also a clear representation of his thinking.

This paper shows that fines can be socially inefficient in a very particular way. I have shown in this paper that fines seem to be effective at direct deterrence: they reduce the specific behavior in locations where citizens know the law is being enforced. Fines seem less capable of generating indirect deterrence: they do nor reduce the specific behavior in locations where citizens know there is no enforcement of the law. This general finding has some nuance when analysing the fines in relation to personal wealth, using relative price of the fixed fine amount. The direct deterrence effect is concentrated on higher quintiles of the wealth distribution, whereas the null effect of indirect deterrence is the combination of a deterrent effect at higher wealth levels and a backfire of fines at lower wealth levels. All of this is consistent with fines being a wake-up call that the citizens receive and only accept if they perceive the fine to be fair or proportional. Becker's logic might need to be complicated in contexts where the State can impose a fine but it can do little to enforce it. It seems clear in the context I study relative price does matter, and is an adittional source of inefficiency since cheaper fines might reduce negative road behavior and increase the amount of money received by the government. This last idea can be supported with existing data on ticket payment, shown in Figure A.24 and Figure A.25.

Also, as I explained before, in the context of Bogotá the only real effect of not paying a speeding fine is the exclusion from the formal market, both in the market for used vehicles and the formal labor market due to the risk of getting a wage garnishment for unpaid fines. Fixed monetary fines might turn into an additional barrier to formality for the poorest members of society. All this point to a clear policy recommendation: wealth based fines. Seeding tickets might have a better effect if they were adjusted, to a lower value, for the less wealthy drivers. If one adds the know fact that higher quintiles have higher payment rates, one can conclude that the State will prevent more speeding, some crashes, and even collect more resources if fines are lower for the poorer. At the same time this might reduce the barrier to formality that existing fines can be. Although not the center of my behavioral exploration, the fact that the desired effects of the policy are concentrated on automoviles, and absent in motorcycles also has important policy implications. This might mean that speeding tickets are not the right tool for the changing face of the public health challenge of road fatalities that some cities are facing, since motorcycles drivers are a growing percentage of road fatalities both as victims and perpetrators. Other interventions are needed if a city like Bogotá (and other in middle income countries) want to face the public health challenge of motorcycles (in all its dimensions: (Martínez-González et al., 2022)). Both non-automated control and behavioral interventions are urgently needed.

I don't think that an implication of the evidence shown is that speed cameras don't work. What this paper has shown is that they do not work through a very specific channel: the behavioral change of marginal speeders just above an unwon real fine threshold. Most evaluations seem to show that in the aggregate Cameras work, and this might be easily explained by their effect on regular speed-limit obeying drivers both internationally (Ang et al., 2020) and specifically in Bogotá (Ferro, 2024). Newer policies might be needed for the selected population that knowing the speed limit and how strongly it is enforced, decides to speed. For the high majority of drivers knowing that there is police enforcement of speeding laws is sufficient to positively influence their behavior.

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Appendix

A Figures

1	158,703	87.44	87.44
2	17,731	9.77	97.21
3	3,460	1.91	99.12
4	934	0.51	99.64
5	371	0.20	99.84
6	142	0.08	99.92
7	52	0.03	99.95
8	39	0.02	99.97
9	21	0.01	99.98
10	10	0.01	99.99
11	13	0.01	99.99
12	3	0.00	99.99
13	6	0.00	100.00
15	1	0.00	100.00
17	1	0.00	100.00
18	1	0.00	100.00
26	1	0.00	100.00
Total	181,489	100.00	

Figure A.1: Times each plate passed exceeding the limit under the Cameras.

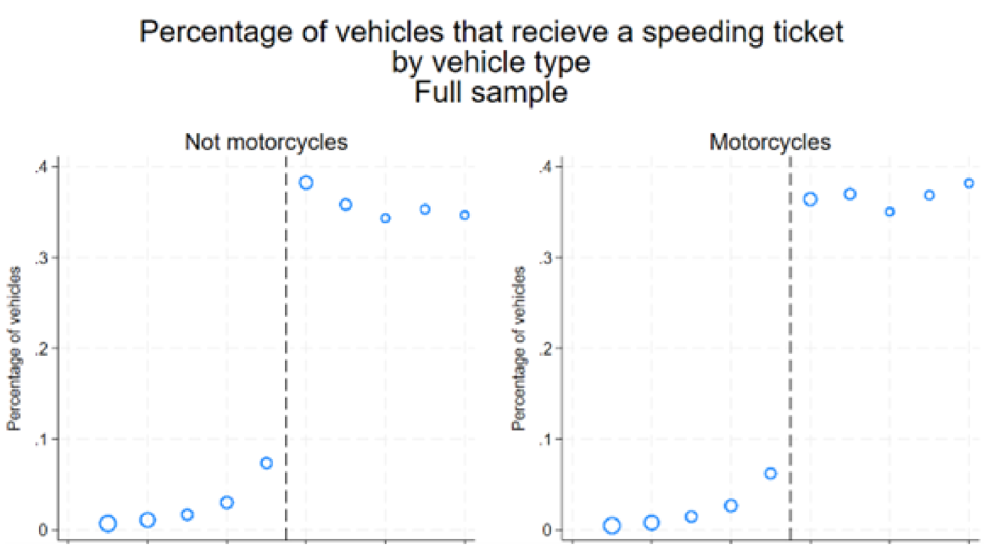

Figure A.2: Existence of the discontinuity by type of vehicle. Full Sample.

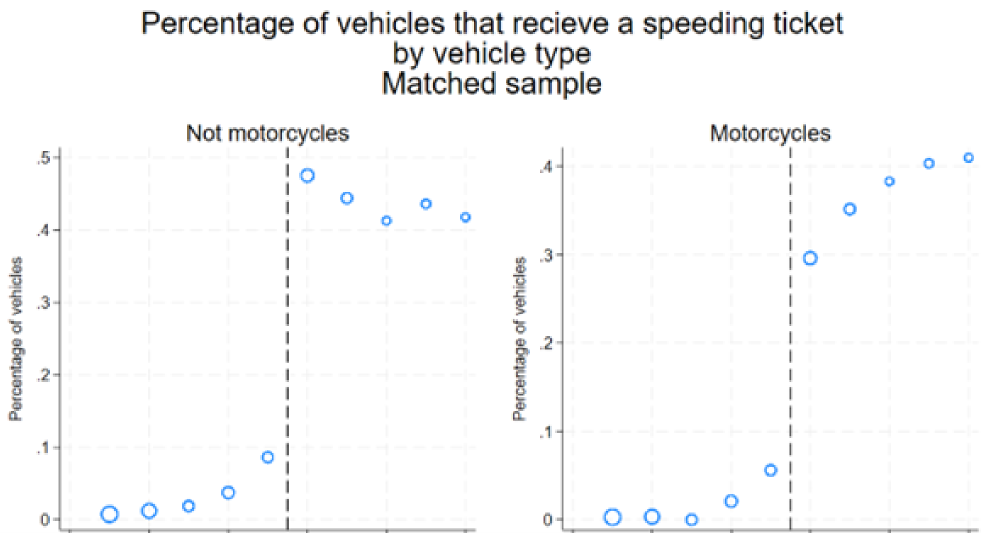

Figure A.3: Existence of the discontinuity by type of vehicle. Matched Sample.

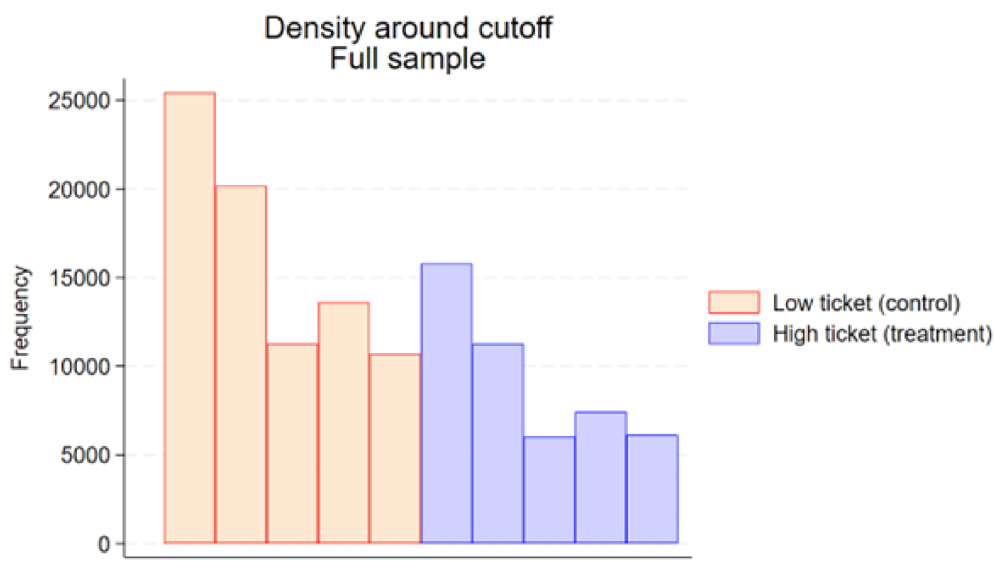

Figure A.4: Frequency by condition. Full sample.

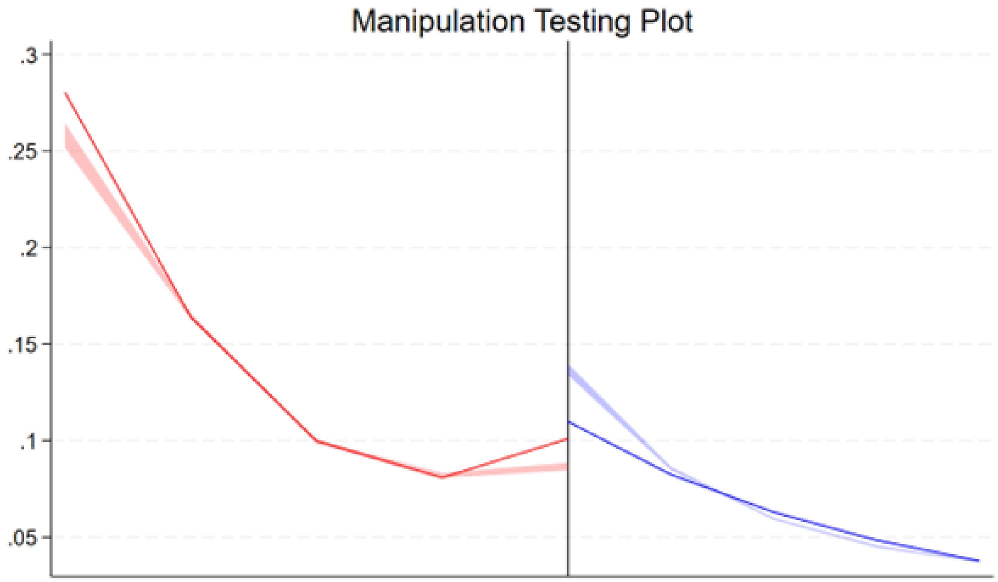

Figure A.5: Manipulation test based on discontinuity. Sample with polynomial of degree 3.

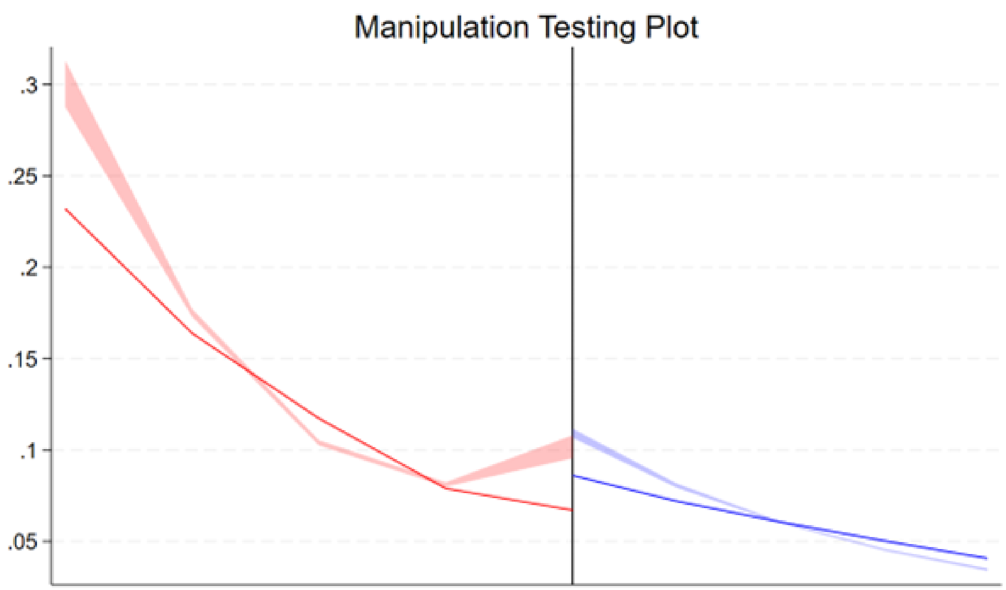

Figure A.6: Manipulation test based on discontinuity. Sample with polynomial of degree 2. Matched Sample.

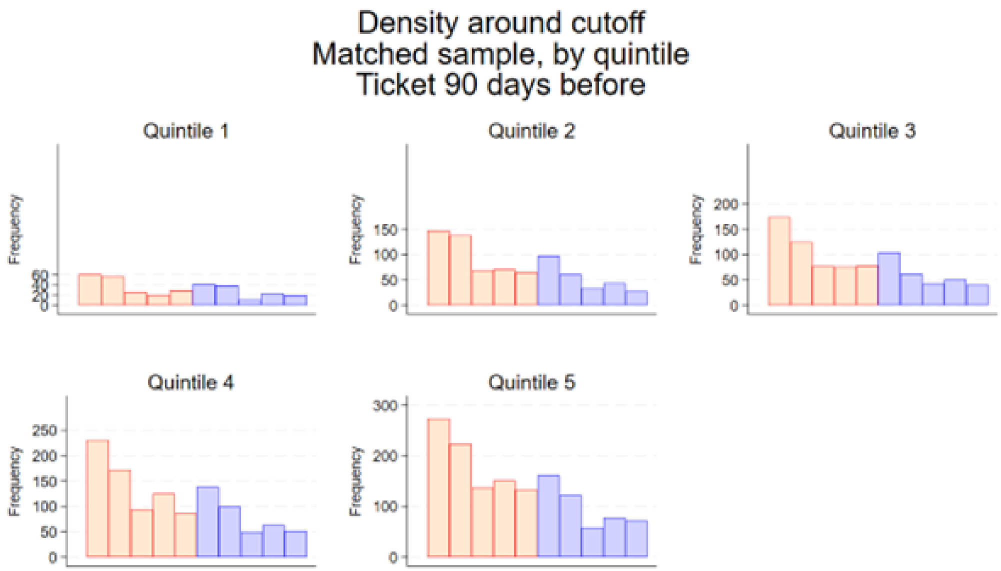

Figure A.7: Frequency by condition and quintile. Matched sample. Vehicles with a previous fine.

Smoothness of observables Matched sample. Quintile 1

	(1) Hour of day	(2) Rush hour dummy	(3) Day of week	(4) Weekend dummy	(5) Motorcycle dummy	(6) Observations per day before	(7) Price of vehicle	(8) Service type	(9) SUV dummy
Bias-corrected	-0.681	-0.147	1.671***	0.439***	-0.699***	-1.078	0.533	0.0365	0.00629
	(1.035)	(0.140)	(0.612)	(0.141)	(0.147)	(0.723)	(1.539)	(0.0357)	(0.0755)
Mean of not ticketed	11.69	0.349	4.219	0.367	0.528	2.304	9.211	0.0237	0.0747
N	7065	7065	7065	7065	7065	7065	7065	7065	7065
N left of cutoff	2692	2692	2692	2692	2692	2692	2692	2692	2692
N right of cutoff	3046	3046	3046	3046	3046	3046	3046	3046	3046
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.8: Smoothness on observables. Quintile 1. Hour is a continuous variable with the time in which the vehicle passed under the camera. Rush hour is a dummy that takes the value of 1 if the vehicle passed under the camera at rush hours (6,7 am or 4,5,6 pm) and 0 otherwise. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess. Type of service takes the value of 1 if a vehicle is used for public service (mainly taxis and similar services) and 0 otherwise. SUV dummy takes the value of 1 if vehicle is an SUV and 0 otherwise.

Smoothness of observables Matched sample. Quintile 2

	(1) Hour of day	(2) Rush hour dummy	(3) Day of week	(4) Weekend dummy	(5) Motorcycle dummy	(6) Observations per day before	(7) Price of vehicle	(8) Service type	(9) SUV dummy
Bias-corrected	-1.206**	-0.0714	0.0334	-0.0445	0.119***	-1.507**	-1.400***	-0.131**	0.0214
	(0.615)	(0.0820)	(0.380)	(0.0862)	(0.0356)	(0.634)	(0.504)	(0.0570)	(0.0459)
Mean of not ticketed	11.87	0.327	4.367	0.415	0.0430	4.478	23.98	0.122	0.0792
N	11849	11849	11849	11849	11849	11849	11849	11849	11849
N left of cutoff	4794	4794	4794	4794	4794	4794	4794	4794	4794
N right of cutoff	4603	4603	4603	4603	4603	4603	4603	4603	4603
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.9: Smoothness on observables. Quintile 2. Hour is a continuous variable with the time in which the vehicle passed under the camera. Rush hour is a dummy that takes the value of 1 if the vehicle passed under the camera at rush hours (6,7 am or 4,5,6 pm) and 0 otherwise. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess. Type of service takes the value of 1 if a vehicle is used for public service (mainly taxis and similar services) and 0 otherwise. SUV dummy takes the value of 1 if vehicle is an SUV and 0 otherwise.

Smoothness of observables Matched sample. Quintile 3

	(1) Hour of day	(2) Rush hour dummy	(3) Day of week	(4) Weekend dummy	(5) Motorcycle dummy	(6) Observations per day before	(7) Price of vehicle	(8) Service type	(9) SUV dummy
Bias-corrected	-0.832	0.0187	-0.237	-0.231**	-0.0391*	2.165***	0.469	-0.196***	0.0512
	(0.727)	(0.0975)	(0.456)	(0.104)	(0.0233)	(0.730)	(0.688)	(0.0695)	(0.0762)
Mean of not ticketed	11.97	0.336	4.405	0.429	0.0138	4.637	34.19	0.127	0.174
N	12823	12823	12823	12823	12823	12823	12823	12823	12823
N left of cutoff	5141	5141	5141	5141	5141	5141	5141	5141	5141
N right of cutoff	5010	5010	5010	5010	5010	5010	5010	5010	5010
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.10: Smoothness on observables. Quintile 3. Hour is a continuous variable with the time in which the vehicle passed under the camera. Rush hour is a dummy that takes the value of 1 if the vehicle passed under the camera at rush hours (6,7 am or 4,5,6 pm) and 0 otherwise. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess. Type of service takes the value of 1 if a vehicle is used for public service (mainly taxis and similar services) and 0 otherwise. SUV dummy takes the value of 1 if vehicle is an SUV and 0 otherwise.

Smoothness of observables Matched sample. Quintile 4

	(1) Hour of day	(2) Rush hour dummy	(3) Day of week	(4) Weekend dummy	(5) Motorcycle dummy	(6) Observations per day before	(7) Price of vehicle	(8) Service type	(9) SUV dummy
Bias-corrected	-0.198	0.0431	0.0177	-0.0852	-0.0456**	0.395	0.320	0.0812	-0.0517
	(0.517)	(0.0703)	(0.326)	(0.0740)	(0.0177)	(0.689)	(1.153)	(0.0586)	(0.0731)
Mean of not ticketed	11.97	0.332	4.352	0.408	0.0114	5.003	51.31	0.190	0.386
N	17225	17225	17225	17225	17225	17225	17225	17225	17225
N left of cutoff	6854	6854	6854	6854	6854	6854	6854	6854	6854
N right of cutoff	6820	6820	6820	6820	6820	6820	6820	6820	6820
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.11: Smoothness on observables. Quintile 4. Hour is a continuous variable with the time in which the vehicle passed under the camera. Rush hour is a dummy that takes the value of 1 if the vehicle passed under the camera at rush hours (6,7 am or 4,5,6 pm) and 0 otherwise. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess. Type of service takes the value of 1 if a vehicle is used for public service (mainly taxis and similar services) and 0 otherwise. SUV dummy takes the value of 1 if vehicle is an SUV and 0 otherwise.

Smoothness of observables Matched sample. Quintile 5

	(1) Hour of day	(2) Rush hour dummy	(3) Day of week	(4) Weekend dummy	(5) Motorcycle dummy	(6) Observations per day before	(7) Price of vehicle	(8) Service type	(9) SUV dummy
Bias-corrected	0.928	-0.317***	0.215	0.0281	0.0000710	-1.084	-12.60	0.0945	-0.121
	(0.575)	(0.0783)	(0.357)	(0.0808)	(0.00906)	(0.941)	(11.83)	(0.0667)	(0.0793)
Mean of not ticketed	12.07	0.340	4.237	0.380	0.00356	5.727	119.1	0.206	0.647
N	18272	18272	18272	18272	18272	18272	18272	18272	18272
N left of cutoff	7068	7068	7068	7068	7068	7068	7068	7068	7068
N right of cutoff	7430	7430	7430	7430	7430	7430	7430	7430	7430
BW left	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.12: Smoothness on observables. Quintile 5. Hour is a continuous variable with the time in which the vehicle passed under the camera. Rush hour is a dummy that takes the value of 1 if the vehicle passed under the camera at rush hours (6,7 am or 4,5,6 pm) and 0 otherwise. Day is a continues variable that goes from 1 (Monday) to 7 (Sunday). Weekend is a dummy that takes the value of 1 for Saturday and Sunday, and 0 otherwise. Motorcycle is a takes the value of 1 for motorcycles, and 0 otherwise. Observations per day before is the total times each vehicle passes under a Camera divided by the number of days before the speed excess. Type of service takes the value of 1 if a vehicle is used for public service (mainly taxis and similar services) and 0 otherwise. SUV dummy takes the value of 1 if vehicle is an SUV and 0 otherwise.

Effect of receiving a ticket. Not motorcyles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.245***	-0.382***	-0.0814***	-0.0923***
	(0.0381)	(0.0848)	(0.0204)	(0.0314)
Mean of not ticketed	0.250	0.418	0.0571	0.0736
N	89037	89037	89037	89037
N Left of cutoff	35012	35012	35012	35012
N Right of cutoff	35827	35827	35827	35827
BW left	4.5	4.5	4.5	4.5
BW right	9.5	9.5	9.5	9.5

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.13: Direct deterrence. Not motorcycles, full sample. Column 1 is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 is a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. Column 4 is a continuous variable, the total of times a vehicle was captured in Major Speeding

Effect of receiving a ticket. Motorcyles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Bias-corrected	-0.0424	-0.123	-0.0341	-0.0407
	(0.0456)	(0.0920)	(0.0274)	(0.0412)
Mean of not ticketed	0.177	0.274	0.0524	0.0678
N	47496	47496	47496	47496
N Left of cutoff	17456	17456	17456	17456
N Right of cutoff	21778	21778	21778	21778
BW left	4.5	4.5	4.5	4.5
BW right	9.5	9.5	9.5	9.5

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.14: Direct deterrence. Motorcycles, full sample. Column 1 is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 is a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. Column 4 is a continuous variable, the total of times a vehicle was captured in Major Speeding

Effect of receiving a ticket

	(1) Below median Speeding >10	(2) Above median Speeding >10	(3) Below median Any deadly crash	(4) Above median Any deadly crash
Bias-corrected	-0.0133	-0.0761***	0.0204***	0.00767***
	(0.0282)	(0.0255)	(0.00524)	(0.00248)
Mean of not ticketed	0.0481	0.0606	0.00161	0.00111
N	24793	42442	24793	42442
Price in dollars	5379	19462	5379	19462
N left of cutoff	9837	16710	9837	16710
N right of cutoff	9937	16974	9937	16974
BW left	4.500	4.500	4.500	4.500
BW left	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.15: Direct and indirect deterrence. Matched sample. Columns 1 and 2 are a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. Columns 3 and 4 are a dummy that takes the value of 1 if the vehicle had at least one crash with a fatality or Triage I attention in the following months, and 0 otherwise.

Effect of receiving a ticket

	(1) Quintile 1 Minor Speeding	(2) Quintile 2 Minor Speeding	(3) Quintile 3 Minor Speeding	(4) Quintile 4 Minor Speeding	(5) Quintile 5 Minor Speeding
Bias-corrected	-0.209	0.0483	0.132	-0.426***	-0.466***
	(0.184)	(0.119)	(0.137)	(0.143)	(0.163)
Mean of not ticketed	0.251	0.370	0.377	0.411	0.466
N	7065	11849	12823	17226	18274
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Rift of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.16: Direct deterrence by quintile of vehicle price. Matched sample Dependent variable is a continuous variable that captures the total times a vehicles is observed between 50 and 60km/h after the first one. For each quintile, variables that are not smooth at cutoff are used as covariates

Effect of receiving a ticket

	(1) Quintile 1 Speeding >10	(2) Quintile 2 Speeding >10	(3) Quintile 3 Speeding >10	(4) Quintile 4 Speeding >10	(5) Quintile 5 Speeding >10
Bias-corrected	0.0263	-0.0841**	-0.0478	-0.0563*	-0.0343
	(0.0635)	(0.0391)	(0.0456)	(0.0341)	(0.0415)
Mean of not ticketed	0.0429	0.0519	0.0513	0.0540	0.0690
N	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Rift of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.17: Direct deterrence by quintile of vehicle price. Matched sample Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. For each quintile, variables that are not smooth at cutoff are used as covariates.

Effect of receiving a ticket

	(1) Quintile 1 Any serious crash	(2) Quintile 2 Any serious crash	(3) Quintile 3 Any serious crash	(4) Quintile 4 Any serious crash	(5) Quintile 5 Any serious crash
Bias-corrected	0.0556***	0.00557	0.0216***	-0.00271	0.0141***
	(0.0172)	(0.00559)	(0.00423)	(0.00171)	(0.00527)
Mean of not ticketed	0.00326	0.000928	0.00117	0.000813	0.00131
N	7065	11849	12823	17226	18272
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Rift of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.18: Indirect deterrence by quintile of vehicle price. Matched sample. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least one deadly crash, and 0 otherwise. For each quintile, variables that are not smooth at cutoff are used as covariates.

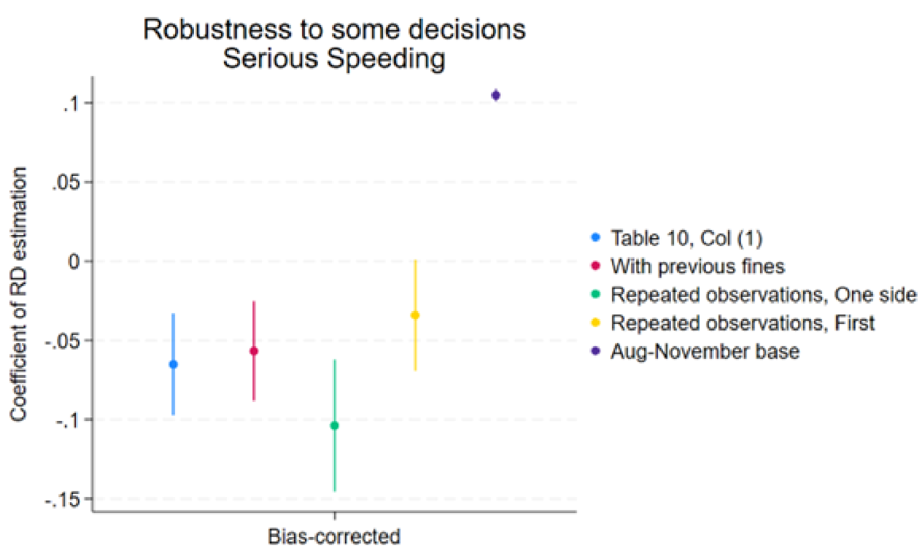

Figure A.18: "Use previous fines" uses the same estimation, this time including vehicles that have had a ticket in the previous six months. "Repeated observations, One Side" shows the coefficient using Vehicles that passed more than once and always above or below the cutoff. "Repeated observations, First" shows the coefficient using Vehicles that passed more than once using the first observations as treatment assignment. "Aug-November" base makes the database bigger, by including all vehicles from August 16 to November 15, 2024.

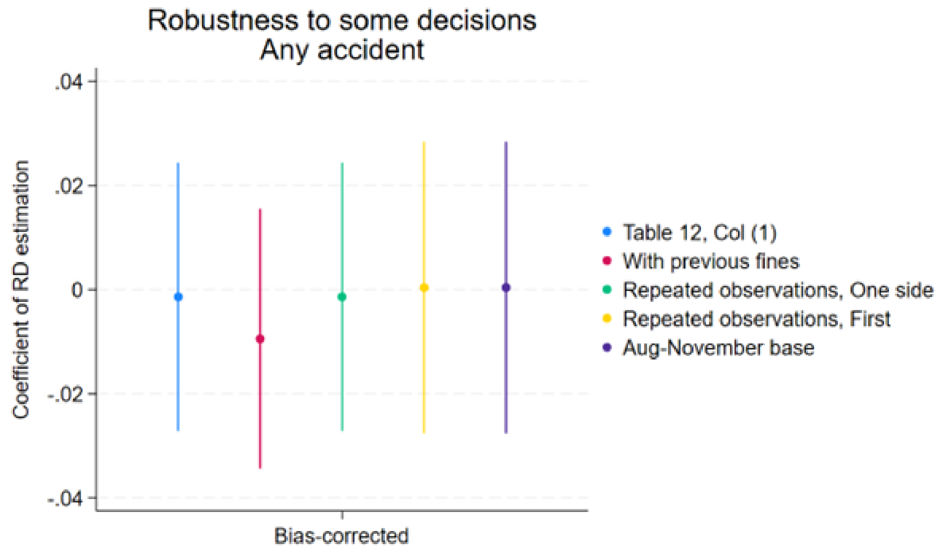

Figure A.19: "Use previous fines" uses the same estimation, this time including vehicles that have had a ticket in the previous six months. "Repeated observations, One Side" shows the coefficient using Vehicles that passed more than once and always above or below the cutoff. "Repeated observations, First" shows the coefficient using Vehicles that passed more than once using the first observations as treatment assignment. "Aug-November" base makes the database bigger, by including all vehicles from August 16 to November 15, 2024.

Effect of agent reviewing ticket. Not motorcycles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Ticket	-0.0446**	-0.103***	-0.0159	-0.0236*
	(0.0185)	(0.0328)	(0.00994)	(0.0138)
Mean of not ticketed	0.179	0.265	0.0451	0.0572
N	94536	94536	94536	94536
First stage Wald-F	65.40	65.40	65.40	65.40

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.20: Direct deterrence. Alternative sample.. Column 1 is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 is a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. Column 4 is a continuous variable, the total of times a vehicle was captured in Major Speeding

Effect of agent reviewing ticket. Motorcycles

	(1) Speeding dummy	(2) Speeding continuous	(3) Speeding >10 dummy	(4) Speeding >10 continuous
Ticket	-0.0235	-0.0109	-0.0273*	-0.0280
	(0.0238)	(0.0393)	(0.0159)	(0.0229)
Mean of not ticketed	0.123	0.176	0.0519	0.0691
N	38132	38132	38132	38132
First stage Wald-F	28.93	28.93	28.93	28.93

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.21: Direct deterrence. Alternative sample.. Column 1 is a dummy that takes the value of 1 if the vehicle had at least one Moderate speeding (between 50 and 60Km/h), and 0 otherwise. Column 2 is a continuous variable, the total of times a vehicle was captured in Moderate speeding. Column 3 is a dummy that takes the value of 1 if the vehicle had at least one Major speeding (above 60 km/h), and 0 otherwise. Column 4 is a continuous variable, the total of times a vehicle was captured in Major Speeding

Effect of agent reviewing ticket. Any crash. By quintile

	(1) Matched Sample	(2) Quintile 1	(3) Quintile 2	(4) Quintile 3	(5) Quintile 4	(6) Quintile 5
Ticket	-0.000286	0.00825	-0.00861	-0.0128	0.00541	0.00699
	(0.00667)	(0.0329)	(0.0162)	(0.00901)	(0.0114)	(0.0142)
Mean of not ticketed	0.0158	0.0330	0.0132	0.0112	0.0114	0.0206
N	48230	3562	8772	10223	12150	13523
First stage Wald-F	46.28	3.551	8.688	13.11	12.64	13.37

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.22: Indirect deterrence by price quintile. Alternative sample.. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least a crash, and 0 otherwise.

Effect of receiving a ticket. Future non-Speeding tickets

	(1) Quintile 1 Other traffic tickets	(2) Quintile 2 Other traffic tickets	(3) Quintile 3 Other traffic tickets	(4) Quintile 4 Other traffic tickets	(5) Quintile 5 Other traffic tickets
Bias-corrected	-0.0205	-0.141**	-0.113*	-0.0520	-0.00661
	(0.101)	(0.0602)	(0.0683)	(0.0467)	(0.0507)
Mean of not ticketed	0.129	0.141	0.120	0.110	0.0974
N	7065	11848	12824	17225	18273
Price in dollars	2295	5999	8553	12845	29482
N Left of cutoff	2692	4794	5141	6854	7067
N Rift of cutoff	3046	4603	5010	6820	7430
BW left	4.500	4.500	4.500	4.500	4.500
BW right	9.500	9.500	9.500	9.500	9.500

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.23: Indirect deterrence by price quintile. Alternative sample.. Dependent variable is a dummy that takes the value of 1 if the vehicle had at least a crash, and 0 otherwise.

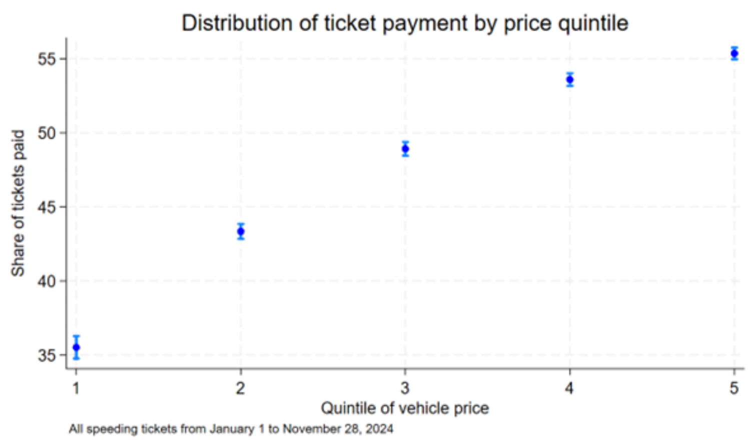

Figure A.24: Ticket payment by quintile of vehicle price. .

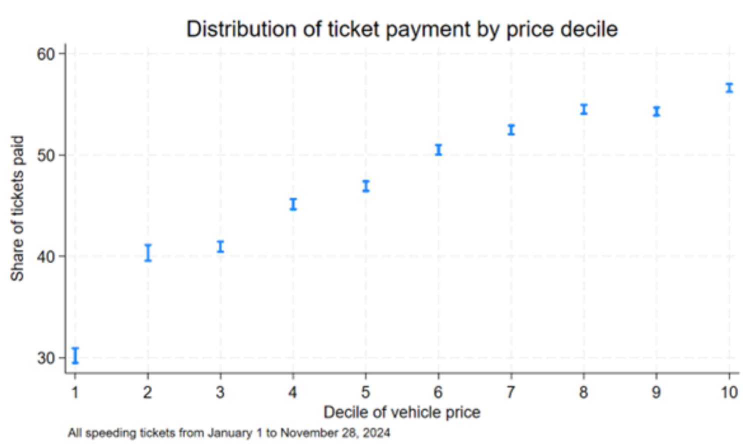

Figure A.25: Ticket payment by decile of vehicle price. .